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MIL-M-28787/507
21 APRIL 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

IOC SEQUENCER MODULE, KEY CODE BWAA

is specification is approved for use by the Naval Sea Systems mmand, Department of the Navy, and is available for use by all partments and Agencies of the Department of Defense.

e requirements for acquiring the modules described herein shall nsist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements d test procedures for the acquisition of an electronic put/Output Controller (IOC) sequencer module.

1.2 Classification. This specification covers two sses of modules. Classes I and II as specified in L-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5963

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 5.799 failures per million operating hours for class I or 14.683 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be BWAA.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 + 5°C 4/ 60 ± 5°C		Class II -55 + 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	Inf	10	Inf	10	Inf	Mohm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---------Module shall pass all remaining tests---------						
D1-D3	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D4	Ibd	V	S	0.0	2.6	0.0	2.6	0.0	2.6	mA
D5	Ibd	V	S	-0.12	4.11	-0.12	4.11	-0.12	4.11	mA
D6-D8	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D9	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D10	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D11-D19	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D20,D21	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D22	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D23	Ibd	V	S	0.0	1.3	0.0	1.3	0.0	1.3	mA
D24-D32	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
E1	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E2,E3	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E4,E5	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E6-E17	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E18-E27	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E28-E33	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E34	Ios	VI	S	-130.0	-30.0	-130.0	-30.0	-130.0	-30.0	mA
E35	Ios	VI	S	-155.0	-65.8	-155.0	-65.8	-155.0	-65.8	mA
E36,E37	Ios	VI	S	-130.0	-30.0	-130.0	-30.0	-130.0	-30.0	mA
E38	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E39	Ios	VI	S	-130.0	-30.0	-130.0	-30.0	-130.0	-30.0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E40	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E41-E56	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E57,E58	Ios	VI	S	-130.0	-30.0	-130.0	-30.0	-130.0	-30.0	mA
F1	Ilh	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F2	Ilh	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F3	Ilh	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F4	Ilh	VII	X	-9.9	0.37	-9.9	0.37	-9.9	0.37	mA
F5	Ilh	VII	X	-6.48	-2.78	-6.48	-2.78	-6.48	-2.78	mA
F6	Ilh	VII	X	-13.4	-2.67	-13.4	-2.67	-13.4	-2.67	mA
F7-F9	Ilh	VII	X	-4.0	0.1	-4 0	0.1	-4.0	0.1	mA
F10	Ilh	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F11	Ilh	VII	X	-0.6	0.04	-0.6	0.04	-0.6	0.04	mA
F12-F19	Ilh	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F20	Ilh	VII	X	-0.8	0.04	-0 8	0.04	-0.8	0.04	mA
F21,F22	Ilh	VII	X	-1.4	0.08	-1.4	0.08	-1.4	0.08	mA
F23	Ilh	VII	X	-0.9	0.04	-0.9	0.04	-0.9	0.04	mA
F24	Ilh	VII	X	-3.8	0.13	-3.8	0.13	-3.8	0.13	mA
F25-F33	Ilh	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G1	Ill	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G2	Ill	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G3	Ill	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G4	Ill	VIII	X	-9.9	0.37	-9.9	0.37	-9.9	0.37	mA
G5	Ill	VIII	X	-6.48	-2.78	-6.48	-2.78	-6.48	-2.78	mA
G6	Ill	VIII	X	-13.4	-2.67	-13.4	-2.67	-13.4	-2.67	mA
G7-G9	Ill	VIII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
G10	Ill	VIII	X	-0.6	0.02	-0 6	0.02	-0.6	0.02	mA
G11	Ill	VIII	X	-0.6	0.04	-0.6	0.04	-0.6	0.04	mA
G12-G19	Ill	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G20	Ill	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G21,G22	Iil	VIII	X	-1.4	0.05	-1.4	0.05	-1.4	0.05	mA
G23	Iil	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G24	Iil	VIII	X	-3.8	0.13	-3.8	0.13	-3.8	0.13	mA
G25-G33	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
H1-H58	Voh	IX	X	2.4	4.5	2 4	4.5	2.4	4.5	V
J1-J59	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
L1	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L2	Vic	XI	S	-1.5	0.0	-1 5	0.0	-1.5	0.0	V
L3-L8	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L9-L18	Vic	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L19,L20	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L21	Vic	XI	S	-1.5	0.0	-1 5	0.0	-1.5	0.0	V
L22	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L23-L31	Vic	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
M1	PSFC	XII	S	4.7	8.8	4.7	8.8	4.7	8.8	uF
M2	PSFC	XII	S	4.3	7.9	4 3	7.9	4.3	7.9	uF
N	Icc	XIII	X	910	2180	910	2180	910	2180	mA
P1,P2	Functional	XIV	X	----------			See table XIV	----------		
Q1,Q2	Dynamic	XV	S	----------			See table XV	----------		
Q3,Q4	Dynamic	XV	Q	----------			See table XV	----------		
R1	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R2	Ci/o	XVII	Q	2	75	2	75	2	75	pF
R3	Ci/o	XVII	Q	2	100	2	100	2	100	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R4	C1/o	XVII	Q	24	321	24	321	24	321	pF
R5	C1/o	XVII	Q	2	175	2	175	2	175	pF
R6	C1/o	XVII	Q	4	100	4	100	4	100	pF
R7,R8	C1/o	XVII	Q	4	75	4	75	4	75	pF
R9	C1/o	XVII	Q	2	75	2	75	2	75	pF
R10	C1/o	XVII	Q	4	75	4	75	4	75	pF
R11-R18	C1/o	XVII	Q	2	50	2	50	2	50	pF
R19	C1/o	XVII	Q	2	75	2	75	2	75	pF
R20,R21	C1/o	XVII	Q	4	75	4	75	4	75	pF
R22	C1/o	XVII	Q	2	75	2	75	2	75	pF
R23	C1/o	XVII	Q	8	100	8	100	8	100	pF
R2-R32	C1/O	XVII	Q	2	50	2	50	2	50	pF
R33,R34	C1/o	XVII	Q	6	200	6	200	6	200	pF
R35-R37	C1/o	XVII	Q	6	150	6	150	6	150	pF
R38	C1/o	XVII	Q	8	175	8	175	8	175	pF
R39	C1/o	XVII	Q	4	75	4	75	4	75	pF
R40,R41	C1/O	XVII	Q	6	100	6	100	6	100	pF
R42	C1/o	XVII	Q	4	100	4	100	4	100	pF
R43	C1/o	XVII	Q	4	75	4	75	4	75	pF
R44,R48	C1/o	XVII	Q	4	100	4	100	4	100	pF
R47-R49	C1/o	XVII	Q	4	75	4	75	4	75	pF
R50	C1/o	XVII	Q	4	100	4	100	4	100	pF
R51-R56	C1/o	XVII	Q	2	50	2	50	2	50	pF
R57-R68	C1/o	XVII	Q	4	75	4	75	4	75	pF
R69	C1/o	XVII	Q	2	50	2	50	2	50	pF
R70	C1/o	XVII	Q	4	75	4	75	4	75	pF
R71,R72	C1/o	XVII	Q	2	50	2	50	2	50	pF
R73	C1/o	XVII	Q	6	200	6	200	6	200	pF
T1,T2	PSTR	XVIII	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/507-1 for class I and M28787/507-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

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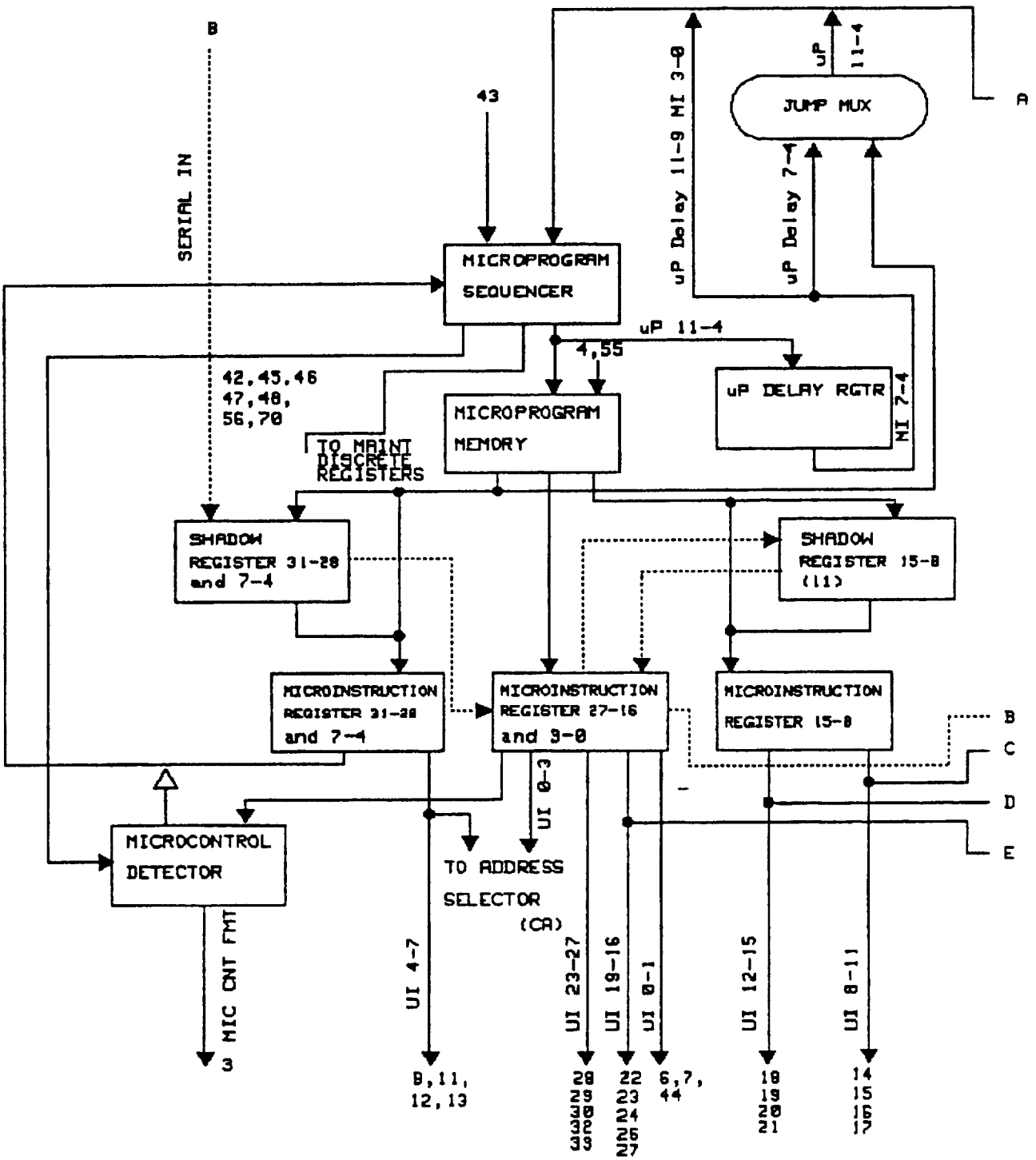


FIGURE 1. BWAA functional block diagram.

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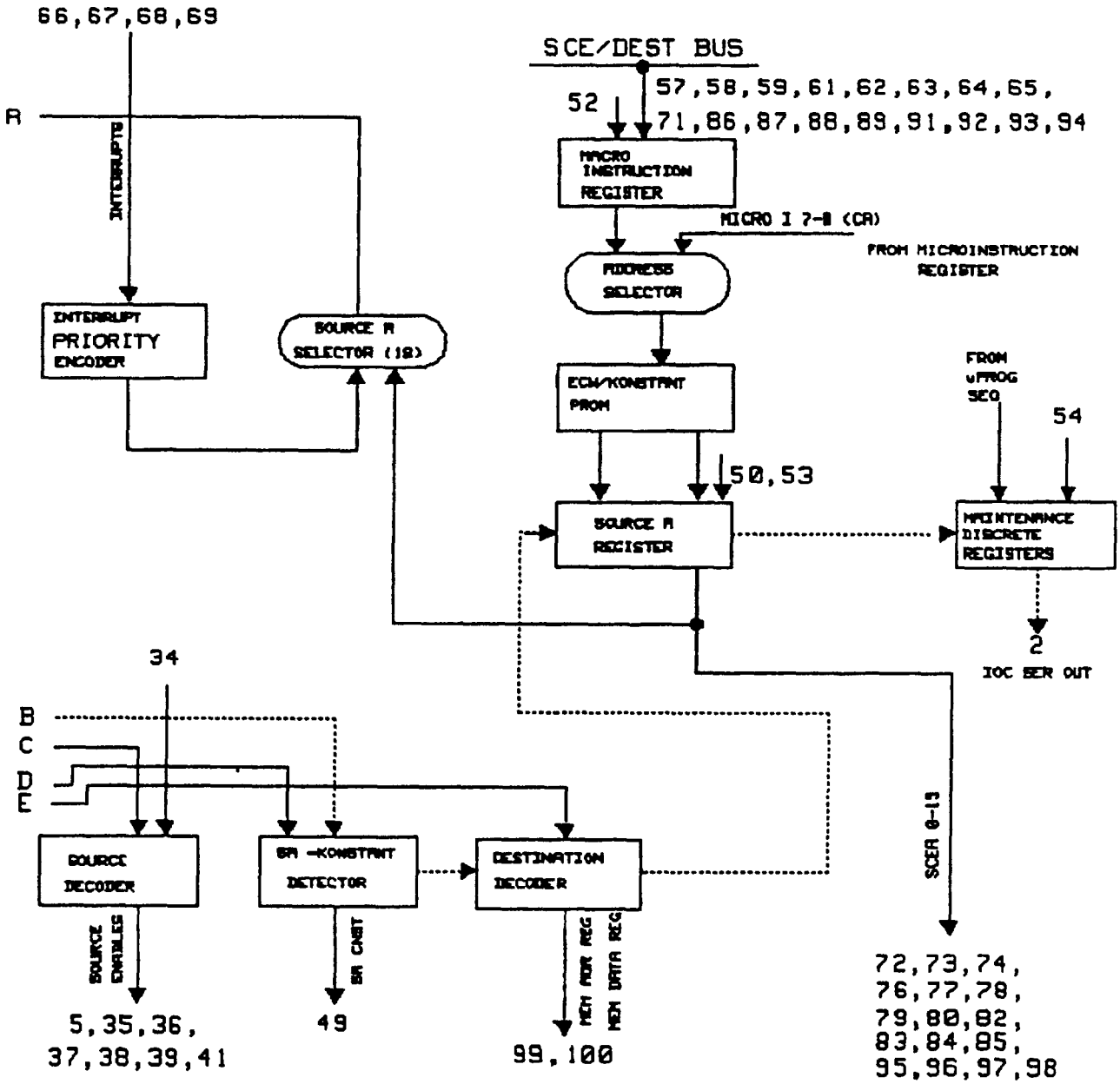


FIGURE 1. BWAA functional block diagram - Continued.

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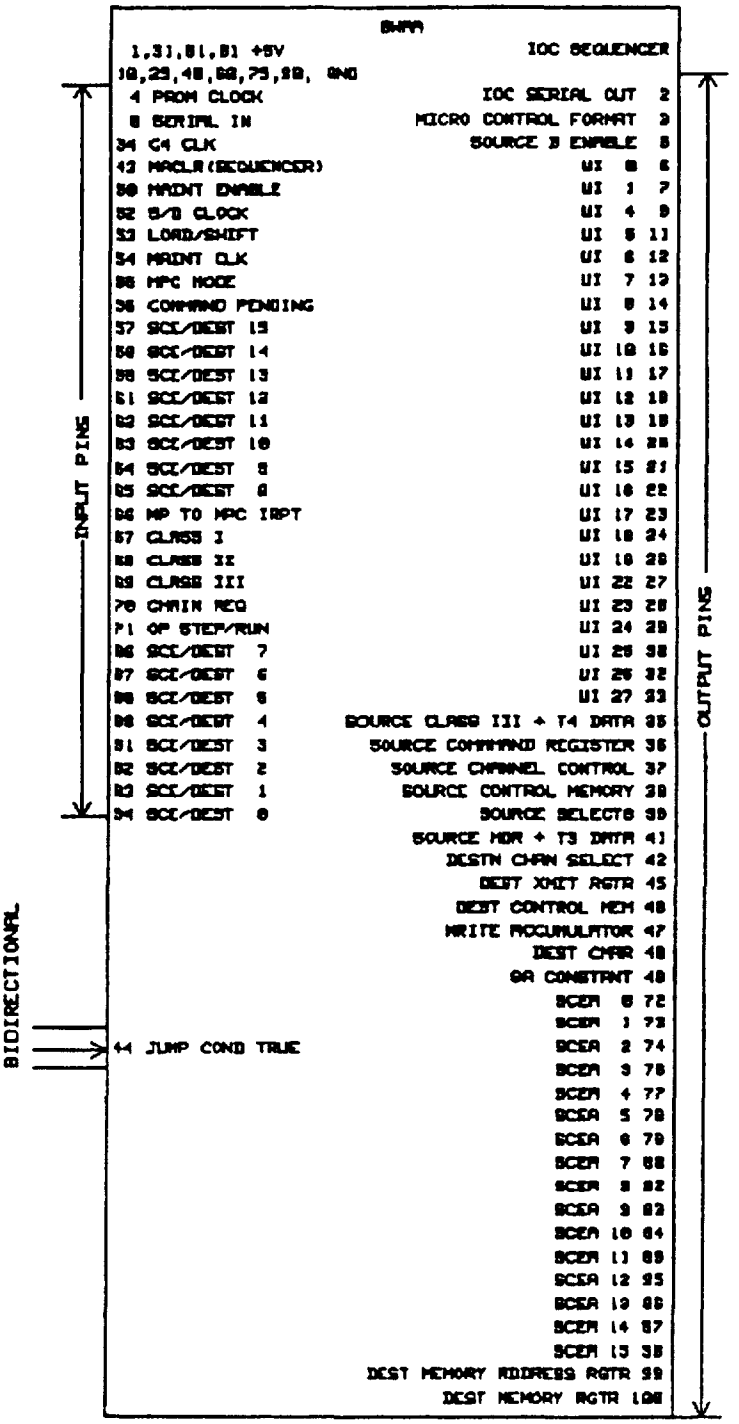


FIGURE 2. BWAA pin assignment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E and L are stress tests and will be performed only at 25 $\pm$ 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E and L where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

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4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

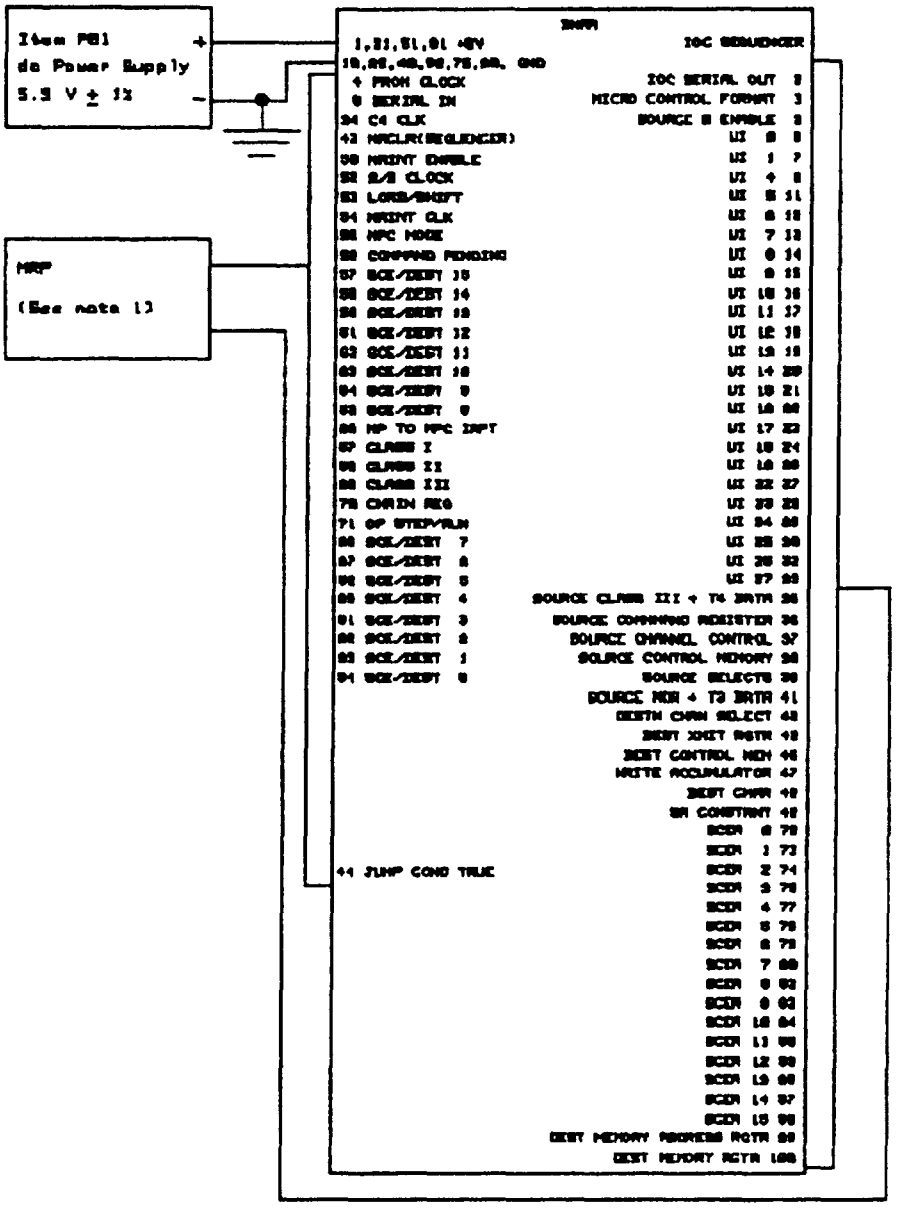
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. BWAA operating temperature and component temperature test circuit.

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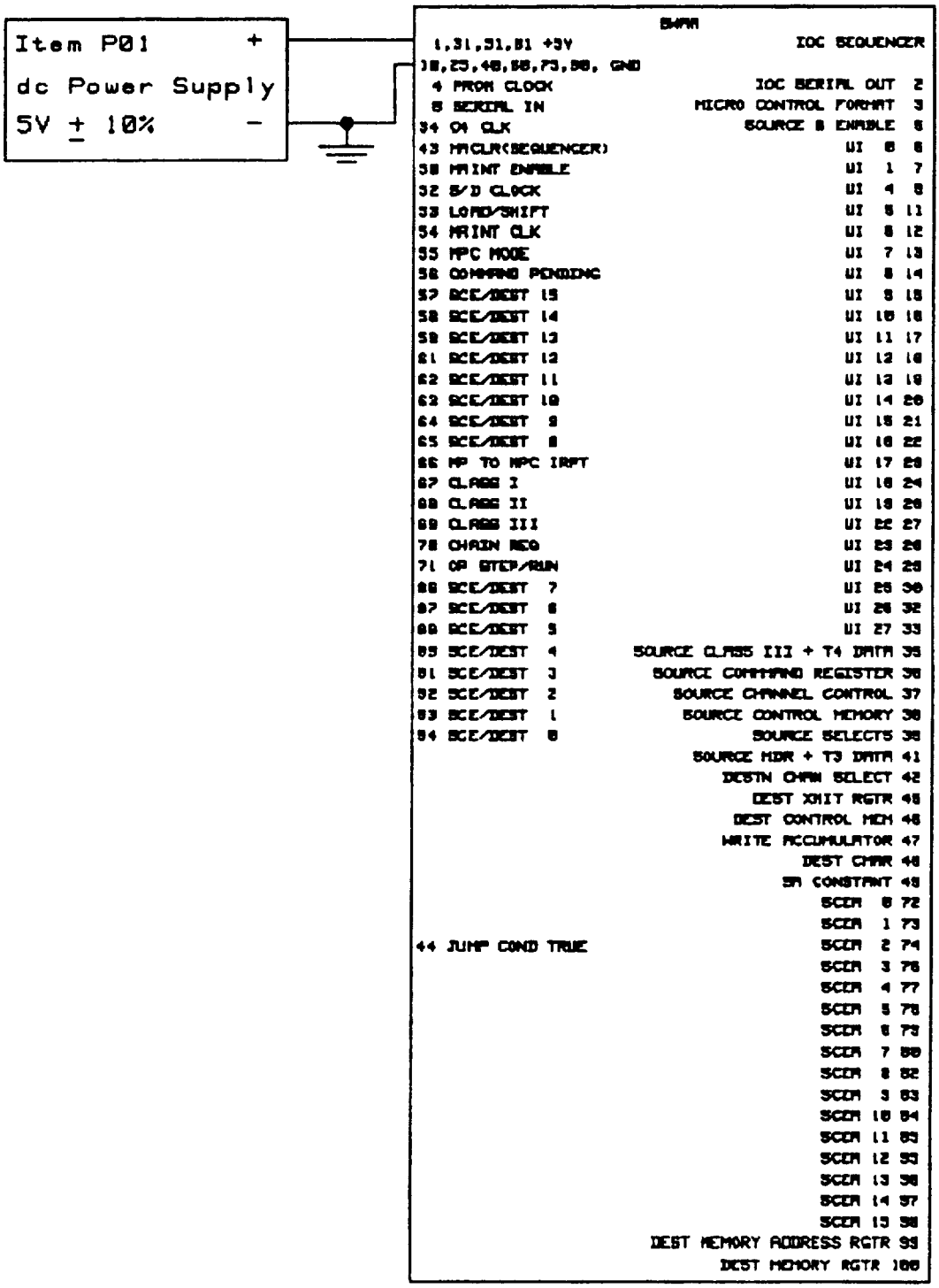
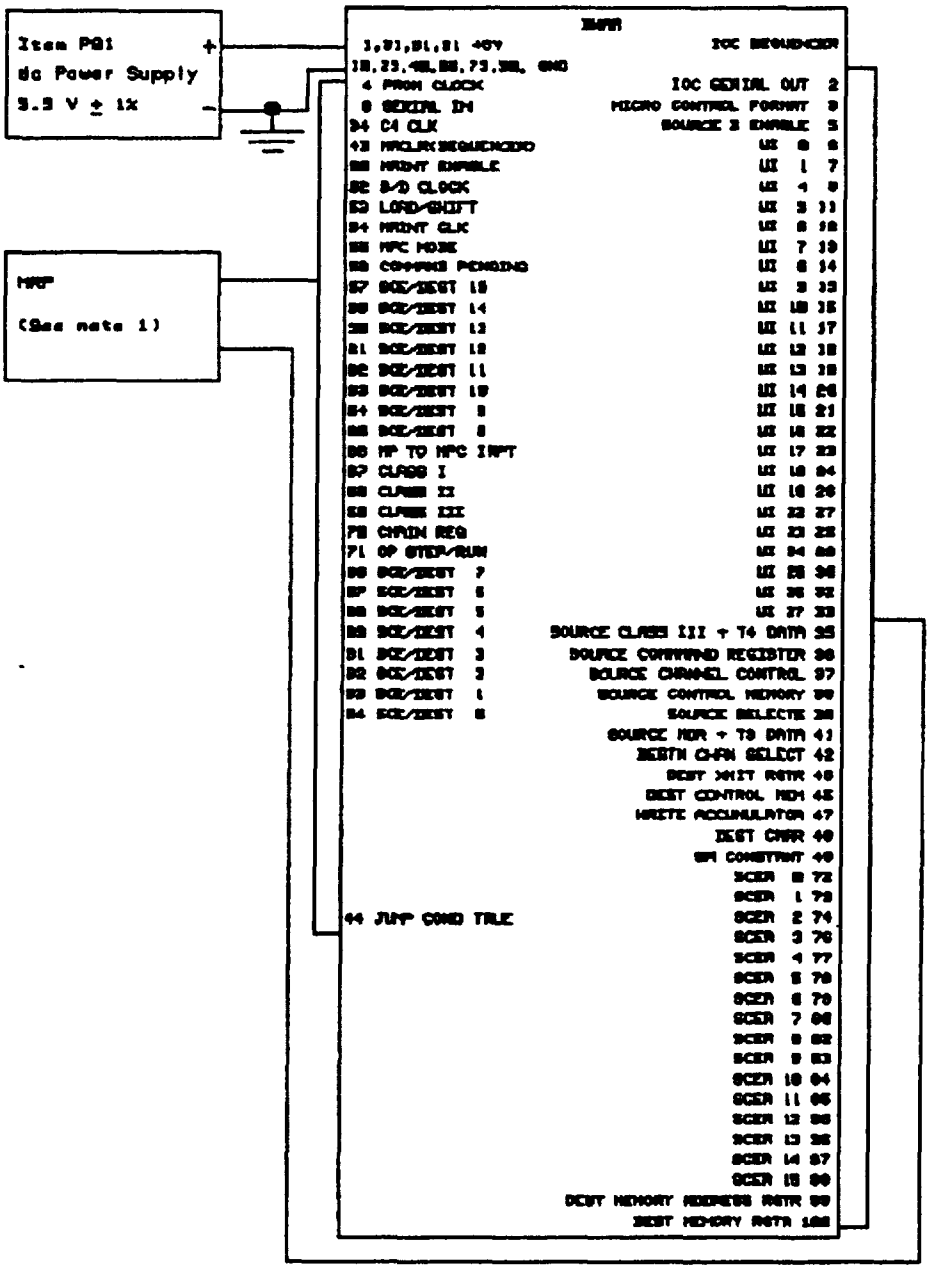


FIGURE 4. BWAA vibration test circuit.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. BWAA life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 11.99 W (max); 8.04 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
PROM CLK	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SERIAL IN	-1.5	0.4	2.4	-0.4	0.02	0.1	75
C4 CLK	-1.2	0.4	2.4	-0.6	0.02	0.1	100
MA CLR	-1.2	0.4	2.4	-9.8	0.37	2.6	321
MAINT ENBL	NA	0.4	2.4	-13.4	-2.67	4.11	175
S/D CLK	-1.2	0.4	2.4	-4.0	0.1	2.0	100
LOAD SHIFT	-1.2	0.4	2.4	-4.0	0.1	2.0	75
MAINT CLK	-1.2	0.4	2.4	-4.0	0.1	2.0	75
MPC MODE	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COMM PNDNG	-1.5	0.4	2.4	-0.6	0.04	0.2	75
SCE/DEST 15	-1.5	0.4	2.4	-0.2	0.02	0.1	50
SCE/DEST 14	-1.5	0.4	2.4	-0.2	0.02	0.1	50
SCE/DEST 13	-1.5	0.4	2.4	-0.2	0.02	0.1	50
SCE/DEST 12	-1.5	0.4	2.4	-0.2	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
SCE/DEST 11	-1.5	0.4	2.4	-0.2	0.02	0.1	50
SCE/DEST 10	-1.5	0.4	2.4	-0.2	0.02	0.1	50
SCE/DEST 9	-1.5	0.4	2.4	-0.2	0.02	0.1	50
SCE/DEST 8	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MP TO MPC	-1.5	0.4	2.4	-0.8	0.04	0.1	75
CL I	-1.2	0.4	2.4	-1.4	0.06	0.2	75
CL II	-1.2	0.4	2.4	-1.4	0.06	0.2	75
CL III	-1.5	0.4	2.4	-0.8	0.04	0.1	75
CHAIN REQ	-1.2	0.4	2.4	-3.8	0.13	1.3	100
OP STP/RN	-1.5	0.4	2.4	-0.2	0.02	0.1	50
SCE/DEST 7	-1.5	0.4	2.4	-0.2	0.02	0.1	50
SCE/DEST 6	-1.5	0.4	2.4	-0.2	0.02	0.1	50
SCE/DEST 5	-1.5	0.4	2.4	-0.2	0.02	0.1	50
SCE/DEST 4	-1.5	0.4	2.4	-0.2	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
SCE/DEST 3	-1.5	0.4	2.4	-0.2	0.02	0.1	50
SCE/DEST 2	-1.5	0.4	2.4	-0.2	0.02	0.1	50
SCE/DEST 1	-1.5	0.4	2.4	-0.2	0.02	0.1	50
SCE/DEST 0	-1.5	0.4	2.4	-0.2	0.02	0.1	50
JMP CON TRU	NA	0.4	2.4	-6.48	-2.78	NA	NA

6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
IOC SER OUT	2.4	0.5	-0.16	1.6	-15.0	NA
MIC CNT FMT	2.4	0.5	-0.4	6.0	-60.0	NA
SC B ENBL	2.4	0.5	-0.4	6.0	-60.0	NA
UI 0	2.4	0.5	-0.16	1.6	-15.0	200
UI 1	2.4	0.5	-0.16	1.6	-15.0	200
UI 4	2.4	0.5	-0.16	1.6	-30.0	150
UI 5	2.4	0.5	-0.16	1.6	-30.0	150

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
UI 6	2.4	0.5	-0.16	1.6	-30.0	150
UI 7	2.4	0.5	-0.16	1.6	-30.0	175
UI 8	2.4	0.5	-0.16	1.6	-30.0	75
UI 9	2.4	0.5	-0.16	1.6	-30.0	100
UI 10	2.4	0.5	-0.16	1.6	-30.0	100
UI 11	2.4	0.5	-0.16	1.6	-30.0	100
UI 12	2.4	0.5	-0.16	1.6	-30.0	75
UI 13	2.4	0.5	-0.16	1.6	-30.0	100
UI 14	2.4	0.5	-0.16	1.6	-30.0	100
UI 15	2.4	0.5	-0.16	1.6	-30.0	100
UI 16	2.4	0.5	-0.16	1.6	-15.0	75
UI 17	2.4	0.5	-0.16	1.6	-15.0	75
UI 18	2.4	0.5	-0.16	1.6	-15.0	75
UI 19	2.4	0.5	-0.16	1.6	-15.0	100

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
UI 22	2.4	0.5	-0.16	1.6	-15.0	50
UI 23	2.4	0.5	-0.16	1.6	-15.0	50
UI 24	2.4	0.5	-0.16	1.6	-15.0	50
UI 25	2.4	0.5	-0.16	1.6	-15.0	50
UI 26	2.4	0.5	-0.16	1.6	-15.0	50
UI 27	2.4	0.5	-0.16	1.6	-15.0	50
C III + T4	2.4	0.5	-0.4	6.0	-60.0	NA
SRC COMM REG	2.4	0.5	-0.4	6.0	-60.0	NA
SRC CHAN CTL	2.4	0.5	-0.4	6.0	-60.0	NA
SRC CTL MEM	2.4	0.5	-0.4	6.0	-60.0	NA
SRC SLCTS	2.4	0.5	-0.4	6.0	-60.0	NA
MDR + T3	2.4	0.5	-0.4	6.0	-60.0	NA
DEST CHN SEL	2.4	0.5	-0.16	1.6	-30.0	NA
JMP CON TRU	2.4	0.5	-0.4	6.0	-65.8	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
DEST XMT REG	2.4	0.5	-0.16	1.6	-30.0	NA
DEST CTL MEM	2.4	0.5	-0.16	1.6	-30.0	NA
WRITE ACC	2.4	0.5	-0.4	6.0	-60.0	NA
DEST CMAR	2.4	0.5	-0.16	1.6	-30.0	NA
SA CNST	2.4	0.5	-0.4	6.0	-60.0	NA
SCEA 0	2.4	0.5	-0.16	1.6	-15.0	75
SCEA 1	2.4	0.5	-0.16	1.6	-15.0	75
SCEA 2	2.4	0.5	-0.16	1.6	-15.0	75
SCEA 3	2.4	0.5	-0.16	1.6	-15.0	75
SCEA 4	2.4	0.5	-0.16	1.6	-15.0	75
SCEA 5	2.4	0.5	-0.16	1.6	-15.0	75
SCEA 6	2.4	0.5	-0.16	1.6	-15.0	75
SCEA 7	2.4	0.5	-0.16	1.6	-15.0	75
SCEA 8	2.4	0.5	-0.16	1.6	-15.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
SCEA 9	2.4	0.5	-0.16	1.6	-15.0	75
SCEA 10	2.4	0.5	-0.16	1.6	-15.0	75
SCEA 11	2.4	0.5	-0.16	1.6	-15.0	75
SCEA 12	2.4	0.5	-0.16	1.6	-15.0	50
SCEA 13	2.4	0.5	-0.16	1.6	-15.0	75
SCEA 14	2.4	0.5	-0.16	1.6	-15.0	50
SCEA 15	2.4	0.5	-0.16	1.6	-15.0	50
MEM ADR REG	2.4	0.5	-0.16	1.6	-30.0	NA
MEM DATA REG	2.4	0.5	-0.16	1.6	-30.0	NA

6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	MAINT CLK to IOC SER OUT	-	-	-	137	137	137	nS	12
1	tpd	MAINT CLK to MIC CNT FMT	-	-	-	137	137	137	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	MAINT CLK to DEST CHN SEL	-	-	-	137	137	137	nS	12
1	tpd	MAINT CLK to JMP CON TRU	-	-	-	137	137	137	nS	12
1	tpd	MAINT CLK to WRITE ACC	-	-	-	137	137	137	nS	12
2	tpd	S/D CLK to UI 0	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 1	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 4	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 5	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 6	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 7	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 8	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 9	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 10	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 11	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 12	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 13	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 14	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 15	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 16	-	-	-	82	82	82	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	S/D CLK to UI 17	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 18	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 19	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 22	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 23	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 24	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 25	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 26	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 27	-	-	-	82	82	82	nS	12
3	tpd	C4 CLK to SC B ENBL	-	-	-	18	18	18	nS	12
4	tpd	MAINT CLK to C III + T4	-	-	-	80	80	80	nS	12
4	tpd	MAINT CLK to SRC COMM REG	-	-	-	80	80	80	nS	12
4	tpd	MAINT CLK to SRC CHAN CTL	-	-	-	80	80	80	nS	12
4	tpd	MAINT CLK to SRC CTL MEM	-	-	-	80	80	80	nS	12
4	tpd	MAINT CLK to SRC SLCTS	-	-	-	80	80	80	nS	12
4	tpd	MAINT CLK to MDR + T3	-	-	-	80	80	80	nS	12

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Check the source to verify that this is the current version before use.

6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	S/D CLK to DEST XMT REG	-	-	-	123	123	123	nS	12
5	tpd	S/D CLK to DEST CTL MEM	-	-	-	123	123	123	nS	12
5	tpd	S/D CLK to DEST CMAR	-	-	-	123	123	123	nS	12
5	tpd	S/D CLK to SA CNST	-	-	-	123	123	123	nS	12
5	tpd	S/D CLK to MEM ADR REG	-	-	-	123	123	123	nS	12
5	tpd	S/D CLK to MEM DATA REG	-	-	-	123	123	123	nS	12
6	tpd	S/D CLK to SCEA 0	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 1	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 2	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 3	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 4	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 5	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 6	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 7	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 8	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 9	-	-	-	81	81	81	nS	12

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Check the source to verify that this is the current version before use.

6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	S/D CLK to SCEA 10	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 11	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 12	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 13	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 14	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 15	-	-	-	81	81	81	nS	12
7	tsu	SERIAL IN to S/D CLK	14	14	14	-	-	-	nS	12
8	th	S/D CLK to SERIAL IN	33	33	33	-	-	-	nS	12
9	tsu	OP STP/RN to S/D CLK	4	4	4	-	-	-	nS	12
10	th	S/D CLK to OP STP/RN	33	33	33	-	-	-	nS	12
11	tsu	SCE DEST 0 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 1 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 2 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 3 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 4 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 5 to S/D CLK	10	10	10	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tsu	SCE DEST 6 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 7 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 8 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 9 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 10 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 11 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 12 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 13 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 14 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 15 to S/D CLK	10	10	10	-	-	-	nS	12

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6.4 Module function. The IOC sequencer module contains the IOC microprogram sequencer, 2048 words of micromemory, microinstruction register, associated control logic to decode the various IOC microinstruction fields, macroinstruction register, Emulation Control Word (ECW) Read-Only Memory (ROM), an interrupt encoder, and Serial Maintenance Interface (SMI) circuits. The IOC sequencer is a microaddress sequencer that determines the source of the next microinstruction to be executed. The sequencer can perform:

(a) Microsubroutine nesting and return (up to five levels).

(b) Conditional or unconditional branching to a given microaddress or an interrupt vector, obtained from the interrupt encoder or ECW ROM.

(c) Microinstruction or microsubroutine looping.

(d) Sequential addressing.

The micromemory is a 2K X 32-bit Programmable Read-Only Memory (PROM) that contains the unalterable IOC microprogram. Microinstruction will be read out of micromemory and placed into the 32-bit microinstruction register. Outputs from the microinstruction register will be supplied to the various inputs of the IOC Arithmetic Logic Unit (ALU). The SMI circuits are part of the serial data path of data/status information used by the maintenance processor for testing the IOC sequencer module.

6.4.1 Microprogram sequencer. The microprogram sequencer provides a 12-bit address used to fetch microinstructions from microprogram memory. The sequencer contains a microprogram counter register, an address register, and a five-word last-in first-out stack. The device can select an address from one of these sources or from an external source (direct inputs). The source is determined by the instruction bits I3-10 (microinstruction bits 31-28). The direct inputs are used to jump to a new location in the microprogram. The microprogram counter register contains the last address plus one and is used to sequentially access microinstructions. The five-word stack is used to execute nested subroutines. Timing for the microprogram sequencer is provided by the destination clock.

6.4.2 Microprogram memory. Microprogram memory is a PROM memory that contains the unalterable IOC microprogram. The PROM contains 2048 32-bit microinstructions.

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6.4.3 Microinstruction register. The 32-bit microinstruction register holds the microinstruction read from microprogram memory.

6.4.4 SMI register. The SMI register holds the IOC sequencer SMI word. The SMI word is 72 bits long, contains the current microinstruction, source A bits and various control signals. The SMI register will be loaded when the load/shift signal is high and maintenance is not enabled. When the maintenance processor enables the SMI, the SMI register is left shifted one bit position on each maintenance processor clock pulse. The data is transferred to/from the maintenance processor via the serial maintenance interface in the bit electronics module.

6.4.5 Source/destination decoders. The source decoder receives its inputs from the microinstruction register bits 10 through 8 (lower three bits of Source B (SB) field). The destination decoder receives microinstruction bits 20, 19 through 16, 7, 6 through 4, and 0 as inputs.

6.4.6 Source A (SA) = konstant decoder. The SA = konstant decoder receives microinstruction bits 15-13 as its input. The decoder is activated by microinstruction with SA = 7. This condition allows a 16-bit constant value to be referenced from the ECW constant PROM as source A.

6.4.7 Instruction register/address selector. The 16-bit instruction register is used to receive the macroinstruction from the source B/Destination bus and transfer it to the address selector for application to the ECW/constant PROM. Addresses will be selected by microinstruction bit 21 or by the select A signal. The macroinstruction is gated into the instruction register by the Input/Output (I/O) clock and the destination instruction register signal.

6.4.8 ECW/constant PROM. The ECW/constant PROM is a 2K by 8-bit PROM that holds the unalterable emulator control words and constants that are used in IOC operation. The PROM is used in two modes; instruction decode and constant look-up. The enable for the PROM includes:

(a) The IOC is in the Microprocessor Controller (MPC) mode.

(b) The IOC sequencer is not being master cleared by the Central Processor (CP) or by a power-up sequence.

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(c) The maintenance panel is not selecting the sequencer and microinstruction bit 21 (constant section reference indicator) is equal to 0 (constant look-up mode).

(d) The sequencer instruction register is loaded with an instruction.

6.4.9 Source A register. The outputs of the constant/ECW PROM are loaded into the source A register whose outputs can be used for jump addresses or to generate constants into the micro structure. The source A register is connected to the ALU source A bus and as an SA source whenever SA = 7. The source A register bits 10-0 can be used as jump addresses for a P = 2 (Jump Memory Address Pointer (JMAP) mnemonic). The source A register will send information to:

(a) The source A bus (bits 15-0).

(b) The source A selector, which selects either the ECW/jump information or interrupt priority information to be applied to the jump address bus and the microprogram sequencer (bits 10-0).

6.4.10 Jump multiplexer. The jump multiplexer receives microinstruction bits 7-0, micro P delay bits 10-4, and micro jump bits 7-0 (source A bits from the source A selector). Outputs from the jump multiplexer are applied to the microprogram sequencer for the next address selection. The jump multiplexer is controlled by the JMAP + ld Memory Data Register (MDR) write signal from both the microprogram sequencer and the destination decoder.

6.4.11 Description of I/O pins.

Signal name	Pin number	I/O	Description
PROM CLK	4	I	PROM CLOCK
SERIAL IN	8	I	SERIAL IN
C4 CLK	34	I	C4 CLK
MA CLR	43	I	MA CLR (SEQUENCER)

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6.4.11 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MAINT ENBL	50	I	MAINT ENABLE
S/D CLK	52	I	S/D CLOCK
LOAD SHIFT	53	I	LOAD SHIFT
MAINT CLK	54	I	MAINT CLK
MPC MODE	55	I	MPC MODE
COMM PNDNG	56	I	COMMAND PENDING
SCE/DEST 15	57	I	SCE/DEST 15
SCE/DEST 14	58	I	SCE/DEST 14
SCE/DEST 13	59	I	SCE/DEST 13
SCE/DEST 12	61	I	SCE/DEST 12
SCE/DEST 11	62	I	SCE/DEST 11
SCE/DEST 10	63	I	SCE/DEST 10
SCE/DEST 9	64	I	SCE/DEST 9
SCE/DEST 8	65	I	SCE/DEST 8
MP TO MPC	66	I	MP TO MPC IRPT
CL I	67	I	CLASS I
CL II	68	I	CLASS II
CL III	69	I	CLASS III
CHAIN REQ	70	I	CHAIN REQ
OP STP/RN	71	I	OP STEP/RUN

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6.4.11 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
SCE/DEST 7	86	I	SCE/DEST 7
SCE/DEST 6	87	I	SCE/DEST 6
SCE/DEST 5	88	I	SCE/DEST 5
SCE/DEST 4	89	I	SCE/DEST 4
SCE/DEST 3	91	I	SCE/DEST 3
SCE/DEST 2	92	I	SCE/DEST 2
SCE/DEST 1	93	I	SCE/DEST 1
SCE/DEST 0	94	I	SCE/DEST 0
IOC SER OUT	2	O	IOC SERIAL OUT
MIC CNT FMT	3	O	MICRO CONTROL FORMAT
SC B ENBL	5	O	SOURCE B ENABLE
UI 0	6	O	UI 0
UI 1	7	O	UI 1
UI 4	9	O	UI 4
UI 5	11	O	UI 5
UI 6	12	O	UI 6
UI 7	13	O	UI 7
UI 8	14	O	UI 8
UI 9	15	O	UI 9
UI 10	16	O	UI 10

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6.4.11 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
UI 11	17	O	UI 11
UI 12	18	O	UI 12
UI 13	19	O	UI 13
UI 14	20	O	UI 14
UI 15	21	O	UI 15
UI 16	22	O	UI 16
UI 17	23	O	UI 17
UI 18	24	O	UI 18
UI 19	26	O	UI 19
UI 22	27	O	UI 22
UI 23	28	O	UI 23
UI 24	29	O	UI 24
UI 25	30	O	UI 25
UI 26	32	O	UI 26
UI 27	33	O	UI 27
C III + T4	35	O	SOURCE CLASS III + T4 DATA
SRC COMM REG	36	O	SOURCE COMMAND REGISTER
SRC CHAN CTL	37	O	SOURCE CHANNEL CONTROL
SRC CTL MEM	38	O	SOURCE CONTROL MEMORY
SRC SLCTS	39	O	SOURCE SELECTS

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6.4.11 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MDR + T3	41	O	SOURCE MDR + T3 DATA
DEST CHN SEL	42	O	DESTN CHAN SELECT
JMP CON TRU	44	I/O	JUMP CONDITION TRUE
DEST XMT REG	45	O	DEST XMIT MEM
DEST CTL MEM	46	O	DEST CONTROL MEM
WRITE ACC	47	O	WRITE ACCUMULATOR
DEST CMAR	48	O	DEST CMAR
SA CNST	49	O	SA CONSTANT
SCEA 0	72	O	SCEA 0
SCEA 1	73	O	SCEA 1
SCEA 2	74	O	SCEA 2
SCEA 3	76	O	SCEA 3
SCEA 4	77	O	SCEA 4
SCEA 5	78	O	SCEA 5
SCEA 6	79	O	SCEA 6
SCEA 7	80	O	SCEA 7
SCEA 8	82	O	SCEA 8
SCEA 9	83	O	SCEA 9
SCEA 10	84	O	SCEA 10
SCEA 11	85	O	SCEA 11

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6.4.11 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
SCEA 12	95	O	SCEA 12
SCEA 13	96	O	SCEA 13
SCEA 14	97	O	SCEA 14
SCEA 15	98	O	SCEA 15
MEM ADR REG	99	O	DEST MEMORY ADDRESS RGTR
MEM DATA REG	100	O	DEST MEMORY DATA RGTR

6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N729)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to	Connect B to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation, frame	C	Frame	10	Inf	Mohm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

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NOTES: - Continued.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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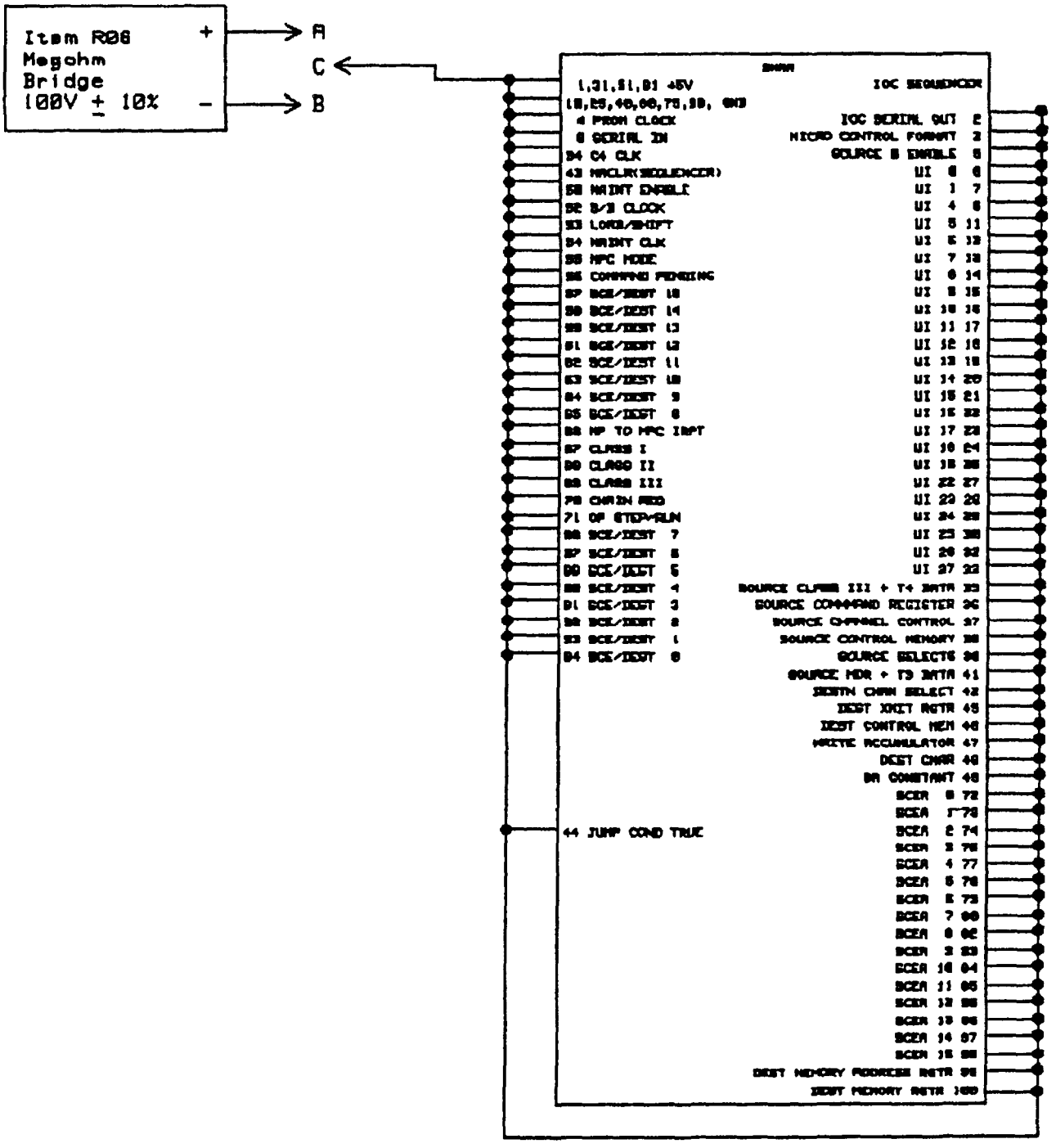


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to	Connect B to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, ground	10	25	0.0	1.0	Ohm	1-5
B2	Continuity, ground	10	40				
B3	Continuity, ground	60	75				
B4	Continuity, ground	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0.0	1.0	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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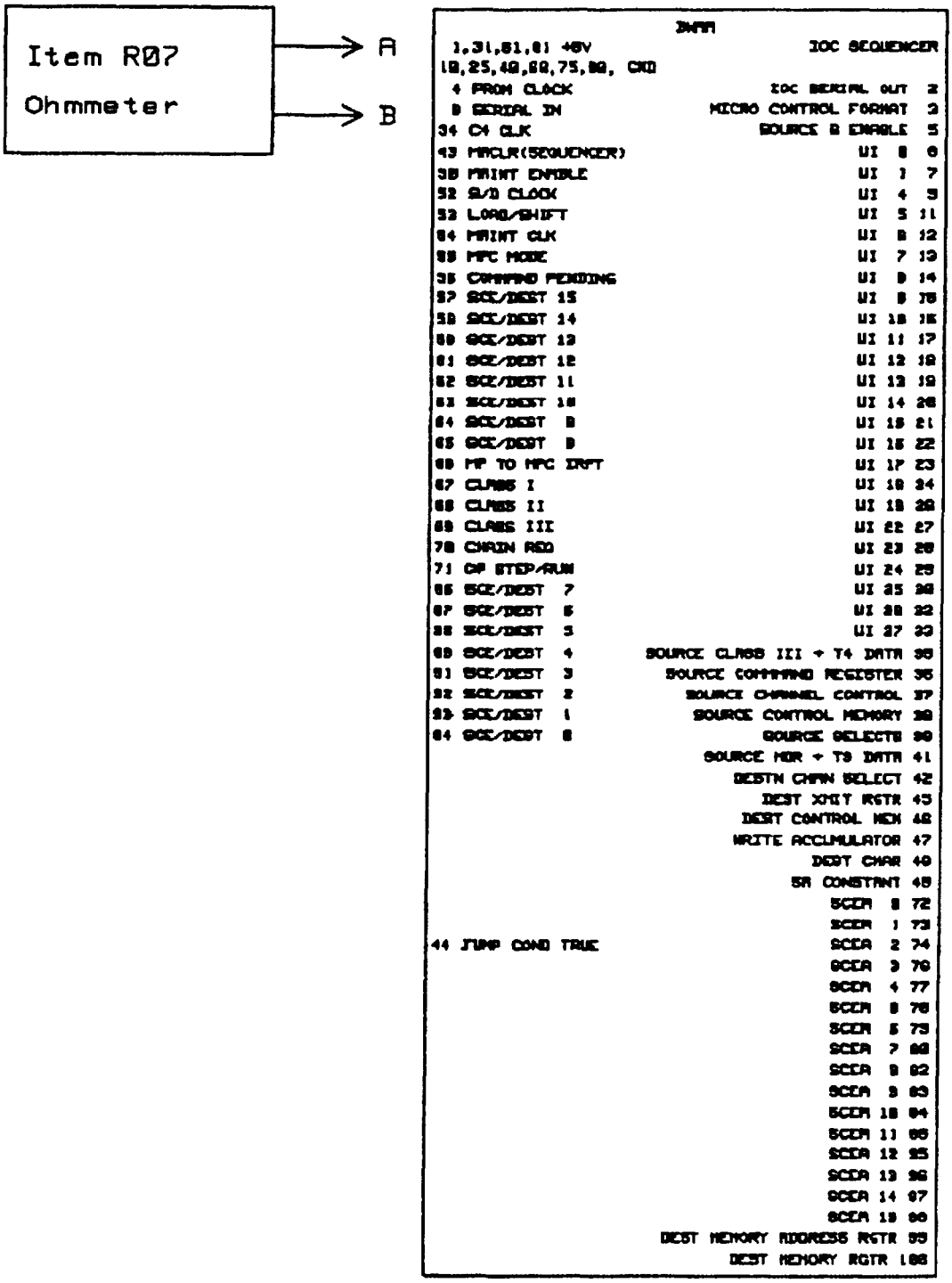


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	3.92 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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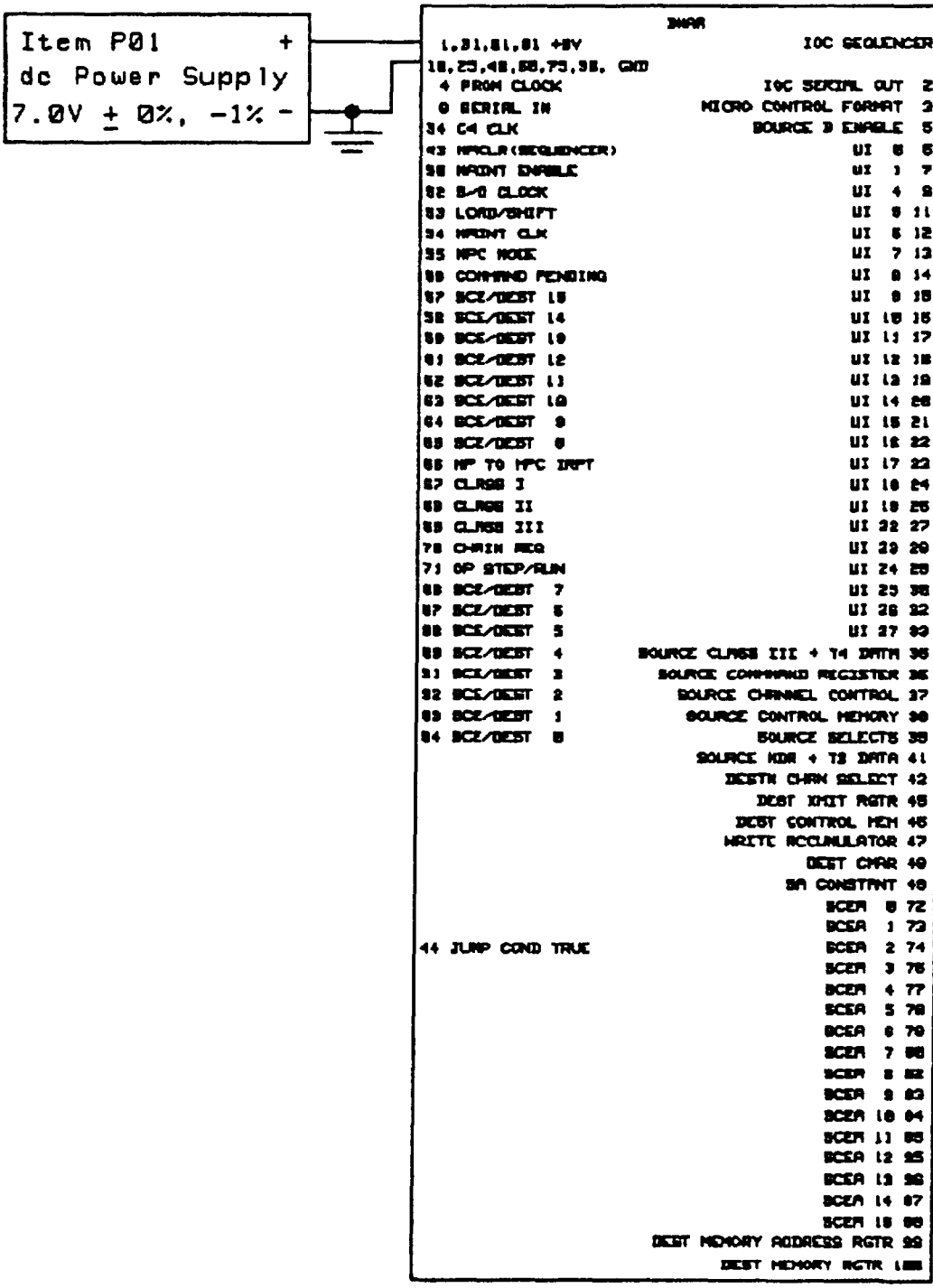


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	PROM CLK	4	5.5	0.0	0.1	mA	1-3
D2	SERIAL IN	8					
D3	C4 CLK	34	5.5	0.0	0.1	mA	1-3
D4	MA CLR	43	5.5	0.0	2.6	mA	1-3
D5	MAINT ENBL	50	5.5	-0.12	4.11	mA	1-3
D6	S/D CLK	52	5.5	0.0	2.0	mA	1-3
D7	LOAD SHIFT	53					
D8	MAINT CLK	54	5.5	0.0	2.0	mA	1-3
D9	MPC MODE	55	5.5	0.0	0.1	mA	1-3
D10	COMM PNDNG	56	5.5	0.0	0.2	mA	1-3
D11	SCE/DEST 15	57	5.5	0.0	0.1	mA	1-3
D12	SCE/DEST 14	58					
D13	SCE/DEST 13	59					
D14	SCE/DEST 12	61					
D15	SCE/DEST 11	62					
D16	SCE/DEST 10	63					
D17	SCE/DEST 9	64					
D18	SCE/DEST 8	65					
D19	MP TO MPC	66	5.5	0.0	0.1	mA	1-3
D20	CL I	67	5.5	0.0	0.2	mA	1-3
D21	CL II	68	5.5	0.0	0.2	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D22	CL III	69	5.5	0.0	0.1	mA	1-3
D23	CHAIN REQ	70	5.5	0.0	1.3	mA	1-3
D24	OP STP/RN	71	5.5	0.0	0.1	mA	1-3
D25	SCE/DEST 7	86					
D26	SCE/DEST 6	87					
D27	SCE/DEST 5	88					
D28	SCE/DEST 4	89					
D29	SCE/DEST 3	91					
D30	SCE/DEST 2	92					
D31	SCE/DEST 1	93					
D32	SCE/DEST 0	94	5.5	0.0	0.1	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D32
Power supply current limit	3.27 A	NA	D1-D32
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D32

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	D1-D32
low, (VL)	0.0 V	± 100 mV	D1-D32
Input current measurement	NA	± 5 uA	D1-D3
	NA	± 130 uA	D4
	NA	± 211 uA	D5
	NA	± 100 uA	D6-D8
	NA	± 5 uA	D9
	NA	± 10 uA	D10
	NA	± 5 uA	D11-D19
	NA	± 10 uA	D20,D21
	NA	± 5 uA	D22
	NA	± 65 uA	D23
	NA	± 5 uA	D24-D32

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	IOC SER OUT	2	0.0	-130.0	-15.0	mA	1-4
E2	MIC CNT FMT	3	0.0	-150.0	-60.0	mA	1-4
E3	SC B ENBL	5	0.0	-150.0	-60.0	mA	1-4
E4	UI 0	6	0.0	-130.0	-15.0	mA	1-4
E5	UI 1	7	0.0	-130.0	-15.0	mA	1-4
E6	UI 4	9	0.0	-260.0	-30.0	mA	1-4
E7	UI 5	11					
E8	UI 6	12					
E9	UI 7	13					
E10	UI 8	14					
E11	UI 9	15					
E12	UI 10	16					
E13	UI 11	17					
E14	UI 12	18					
E15	UI 13	19					
E16	UI 14	20					
E17	UI 15	21	0.0	-260.0	-30.0	mA	1-4
E18	UI 16	22	0.0	-130.0	-15.0	mA	1-4
E19	UI 17	23					
E20	UI 18	24					
E21	UI 19	26	0.0	-130.0	-15.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E22	UI 22	27	0.0	-130.0	-15.0	mA	1-4
E23	UI 23	28					
E24	UI 24	29					
E25	UI 25	30					
E26	UI 26	32					
E27	UI 27	33	0.0	-130.0	-15.0	mA	1-4
E28	C III + T4	35	0.0	-150.0	-60.0	mA	1-4
E29	SRC COMM REG	36					
E30	SRC CHAN CTL	37					
E31	SRC CTL MEM	38					
E32	SRC SLCTS	39					
E33	MDR + T3	41	0.0	-150.0	-60.0	mA	1-4
E34	DEST CHN SEL	42	0.0	-130.0	-30.0	mA	1-4
E35	JMP CON TRU	44	0.0	-155.0	-65.8	mA	1-4
E36	DEST XMT REG	45	0.0	-130.0	-30.0	mA	1-4
E37	DEST CTL MEM	46	0.0	-130.0	-30.0	mA	1-4
E38	WRITE ACC	47	0.0	-150.0	-60.0	mA	1-4
E39	DEST CMAR	48	0.0	-130.0	-30.0	mA	1-4
E40	SA CNST	49	0.0	-150.0	-60.0	mA	1-4
E41	SCEA 0	72	0.0	-130.0	-15.0	mA	1-4
E42	SCEA 1	73	0.0	-130.0	-15.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E43	SCEA 2	74	0.0	-130.0	-15.0	mA	1-4
E44	SCEA 3	76					
E45	SCEA 4	77					
E46	SCEA 5	78					
E47	SCEA 6	79					
E48	SCEA 7	80					
E49	SCEA 8	82					
E50	SCEA 9	83					
E51	SCEA 10	84					
E52	SCEA 11	85					
E53	SCEA 12	95					
E54	SCEA 13	96					
E55	SCEA 14	97					
E56	SCEA 15	98	0.0	-130.0	-15.0	mA	1-4
E57	MEM ADR REG	99	0.0	-130.0	-30.0	mA	1-4
E58	MEM DATA REG	100	0.0	-130.0	-30.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E58
Power supply current limit	3.27 A	NA	E1-E58
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E58
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E58 E1-E58
Output current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 5.75 mA ± 4.50 mA ± 5.75 mA ± 11.5 mA ± 5.75 mA ± 4.50 mA ± 5.0 mA ± 4.46 mA ± 5.0 mA ± 4.50 mA ± 5.0 mA ± 4.50 mA ± 5.75 mA ± 5.0 mA	E1 E2,E3 E4,E5 E6-E17 E18-E27 E28-E33 E34 E35 E36,E37 E38 E39 E40 E41-E56 E57,E58

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	PROM CLK	4	2.4	-0.6	0.02	mA	1-3
F2	SERIAL IN	8	2.4	-0.4	0.02	mA	1-3
F3	C4 CLK	34	2.4	-0.6	0.02	mA	1-3
F4	MA CLR	43	2.4	-9.8	0.37	mA	1-3
F5	JMP CON TRU	44	2.4	-6.48	-2.78	mA	1-3
F6	MAINT ENBL	50	2.4	-13.4	-2.67	mA	1-3
F7	S/D CLK	52	2.4	-4.0	0.1	mA	1-3
F8	LOAD SHIFT	53					
F9	MAINT CLK	54	2.4	-4.0	0.1	mA	1-3
F10	MPC MODE	55	2.4	-0.6	0.02	mA	1-3
F11	COMM PNDNG	56	2.4	-0.6	0.04	mA	1-3
F12	SCE/DEST 15	57	2.4	-0.2	0.02	mA	1-3
F13	SCE/DEST 14	58					
F14	SCE/DEST 13	59					
F15	SCE/DEST 12	61					
F16	SCE/DEST 11	62					
F17	SCE/DEST 10	63					
F18	SCE/DEST 9	64					
F19	SCE/DEST 8	65	2.4	-0.2	0.02	mA	1-3
F20	MP TO MPC	66	2.4	-0.8	0.04	mA	1-3
F21	CL I	67	2.4	-1.4	0.06	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F22	CL II	68	2.4	-1.4	0.06	mA	1-3
F23	CL III	69	2.4	-0.8	0.04	mA	1-3
F24	CHAIN REQ	70	2.4	-3.8	0.13	mA	1-3
F25	OP STP/RN	71	2.4	-0.2	0.02	mA	1-3
F26	SCE/DEST 7	86					
F27	SCE/DEST 6	87					
F28	SCE/DEST 5	88					
F29	SCE/DEST 4	89					
F30	SCE/DEST 3	91					
F31	SCE/DEST 2	92					
F32	SCE/DEST 1	93					
F33	SCE/DEST 0	94	2.4	-0.2	0.02	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F33
Power supply current limit	3.27 A	NA	F1-F33

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F33
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F33
low, (VL)	0.0 V	± 100 mV	F1-F33
Input current measurement	NA	± 31 uA	F1
	NA	± 21 uA	F2
	NA	± 31 uA	F3
	NA	± 510 uA	F4
	NA	± 185 uA	F5
	NA	± 540 uA	F6
	NA	± 205 uA	F7-F9
	NA	± 31 uA	F10
	NA	± 32 uA	F11
	NA	± 11 uA	F12-F19
	NA	± 42 uA	F20
	NA	± 73 uA	F21,F22
	NA	± 42 uA	F23
	NA	± 196 uA	F24
	NA	± 11 uA	F25-F33

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	PROM CLK	4	0.4	-0.6	0.02	mA	1-3
G2	SERIAL IN	8	0.4	-0.4	0.02	mA	1-3
G3	C4 CLK	34	0.4	-0.6	0.02	mA	1-3
G4	MA CLR	43	0.4	-9.8	0.37	mA	1-3
G5	JMP CON TRU	44	0.4	-6.48	-2.78	mA	1-3
G6	MAINT ENBL	50	0.4	-13.4	-2.67	mA	1-3
G7	S/D CLK	52	0.4	-4.0	0.1	mA	1-3
G8	LOAD SHIFT	53					
G9	MAINT CLK	54	0.4	-4.0	0.1	mA	1-3
G10	MPC MODE	55	0.4	-0.6	0.02	mA	1-3
G11	COMM PNDNG	56	0.4	-0.6	0.04	mA	1-3
G12	SCE/DEST 15	57	0.4	-0.2	0.02	mA	1-3
G13	SCE/DEST 14	58					
G14	SCE/DEST 13	59					
G15	SCE/DEST 12	61					
G16	SCE/DEST 11	62					
G17	SCE/DEST 10	63					
G18	SCE/DEST 9	64					
G19	SCE/DEST 8	65	0.4	-0.2	0.02	mA	1-3
G20	MP TO MPC	66	0.4	-0.8	0.04	mA	1-3
G21	CL I	67	0.4	-1.4	0.06	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G22	CL II	68	0.4	-1.4	0.06	mA	1-3
G23	CL III	69	0.4	-0.8	0.04	mA	1-3
G24	CHAIN REQ	70	0.4	-3.8	0.13	mA	1-3
G25	OP STP/RN	71	0.4	-0.2	0.02	mA	1-3
G26	SCE/DEST 7	86					
G27	SCE/DEST 6	87					
G28	SCE/DEST 5	88					
G29	SCE/DEST 4	89					
G30	SCE/DEST 3	91					
G31	SCE/DEST 2	92					
G32	SCE/DEST 1	93					
G33	SCE/DEST 0	94	0.4	-0.2	0.02	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G33
Power supply current limit	3.27 A	NA	G1-G33

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G33
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	G1-G33
low, (VL)	0.0 V	± 100 mV	G1-G33
Input current measurement	NA	± 31 uA	G1
	NA	± 21 uA	G2
	NA	± 31 uA	G3
	NA	± 510 uA	G4
	NA	± 185 uA	G5
	NA	± 540 uA	G6
	NA	± 205 uA	G7-G9
	NA	± 31 uA	G10
	NA	± 32 uA	G11
	NA	± 11 uA	G12-G19
	NA	± 42 uA	G20
	NA	± 73 uA	G21,G22
	NA	± 42 uA	G23
	NA	± 196 uA	G24
	NA	± 11 uA	G25-G33

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	IOC SER OUT	2	-0.16	2.4	4.5	V	1-4
H2	MIC CNT FMT	3	-0.4	2.4	4.5	V	1-4
H3	SC B ENBL	5	-0.4	2.4	4.5	V	1-4
H4	UI 0	6	-0.16	2.4	4.5	V	1-4
H5	UI 1	7					
H6	UI 4	9					
H7	UI 5	11					
H8	UI 6	12					
H9	UI 7	13					
H10	UI 8	14					
H11	UI 9	15					
H12	UI 10	16					
H13	UI 11	17					
H14	UI 12	18					
H15	UI 13	19					
H16	UI 14	20					
H17	UI 15	21					
H18	UI 16	22					
H19	UI 17	23					
H20	UI 18	24					
H21	UI 19	26	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H22	UI 22	27	-0.16	2.4	4.5	V	1-4
H23	UI 23	28					
H24	UI 24	29					
H25	UI 25	30					
H26	UI 26	32					
H27	UI 27	33	-0.16	2.4	4.5	V	1-4
H28	C III + T4	35	-0.4	2.4	4.5	V	1-4
H29	SRC COMM REG	36					
H30	SRC CHAN CTL	37					
H31	SRC CTL MEM	38					
H32	SRC SLCTS	39					
H33	MDR + T3	41	-0.4	2.4	4.5	V	1-4
H34	DEST CHN SEL	42	-0.16	2.4	4.5	V	1-4
H35	JMP CON TRU	44	-0.4	2.4	4.5	V	1-4
H36	DEST XMT REG	45	-0.16	2.4	4.5	V	1-4
H37	DEST CTL MEM	46	-0.16	2.4	4.5	V	1-4
H38	WRITE ACC	47	-0.4	2.4	4.5	V	1-4
H39	DEST CMAR	48	-0.16	2.4	4.5	V	1-4
H40	SA CNST	49	-0.4	2.4	4.5	V	1-4
H41	SCEA 0	72	-0.16	2.4	4.5	V	1-4
H42	SCEA 1	73	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H43	SCEA 2	74	-0.16	2.4	4.5	V	1-4
H44	SCEA 3	76					
H45	SCEA 4	77					
H46	SCEA 5	78					
H47	SCEA 6	79					
H48	SCEA 7	80					
H49	SCEA 8	82					
H50	SCEA 9	83					
H51	SCEA 10	84					
H52	SCEA 11	85					
H53	SCEA 12	95					
H54	SCEA 13	96					
H55	SCEA 14	97					
H56	SCEA 15	98					
H57	MEM ADR REG	99					
H58	MEM DATA REG	100	-0.16	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H58
Power supply current limit	3.27 A	NA	H1-H58
Forcing current, (Ioh)	-0.16 mA	± 16 uA	H1
	-0.4 mA	± 40 uA	H2,H3
	-0.16 mA	± 16 uA	H4-H27
	-0.4 mA	± 40 uA	H28-H33
	-0.16 mA	± 16 uA	H34
	-0.4 mA	± 40 uA	H35
	-0.16 mA	± 16 uA	H36,H37
	-0.4 mA	± 40 uA	H38
	-0.16 mA	± 16 uA	H39
	-0.4 mA	± 40 uA	H40
	-0.16 mA	± 16 uA	H41-H58
Pattern voltage high, (VH)	3.0 V	± 100 mV	H1-H58
low, (VL)	0.0 V	± 100 mV	H1-H58
Output voltage measurement	NA	± 105 mV	H1-H58

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

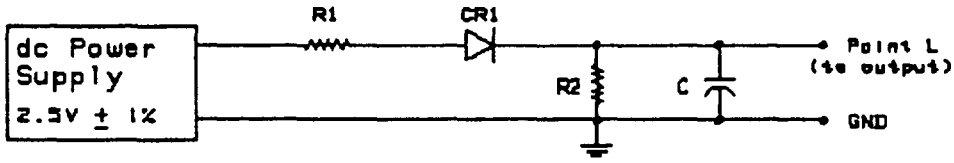
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current follow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,6,7,9,11,12,13, 14,15,16,17,18,19, 20,21,22,23,24,26, 27,28,29,30,32,33, 42,45,46,48,72,73, 74,76,77,78,79,80, 82,83,84,85,95,96, 97,98,99,100	16000	1.6	0.5
3,5,35,36,37,38, 39,41,44,47,49	6200	6.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	IOC SER OUT	2	1.6	0.0	0.5	V	1-4
J2	MIC CNT FMT	3	6.0	0.0	0.5	V	1-4
J3	SC B ENBL	5	6.0	0.0	0.5	V	1-4
J4	UI 0	6	1.6	0.0	0.5	V	1-4
J5	UI 1	7					
J6	UI 4	9					
J7	UI 5	11					
J8	UI 6	12					
J9	UI 7	13					
J10	UI 8	14					
J11	UI 9	15					
J12	UI 10	16					
J13	UI 11	17					
J14	UI 12	18					
J15	UI 13	19					
J16	UI 14	20					
J17	UI 15	21					
J18	UI 16	22					
J19	UI 17	23					
J20	UI 18	24					
J21	UI 19	26	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J22	UI 22	27	1.6	0.0	0.5	V	1-4
J23	UI 23	28					
J24	UI 24	29					
J25	UI 25	30					
J26	UI 26	32					
J27	UI 27	33	1.6	0.0	0.5	V	1-4
J28	C III + T4	35	6.0	0.0	0.5	V	1-4
J29	SRC COMM REG	36					
J30	SRC CHAN CTL	37					
J31	SRC CTL MEM	38					
J32	SRC SLCTS	39					
J33	MDR + T3	41	6.0	0.0	0.5	V	1-4
J34	DEST CHN SEL	42	1.6	0.0	0.5	V	1-4
J35	JMP CON TRU	44	6.0	0.0	0.5	V	1-4
J36	DEST XMT REG	45	1.6	0.0	0.5	V	1-4
J37	DEST CTL MEM	46	1.6	0.0	0.5	V	1-4
J38	WRITE ACC	47	6.0	0.0	0.5	V	1-4
J39	DEST CMAR	48	1.6	0.0	0.5	V	1-4
J40	SA CNST	49	6.0	0.0	0.5	V	1-4
J41	SCEA 0	72	1.6	0.0	0.5	V	1-4
J42	SCEA 1	73	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J43	SCEA 2	74	1.6	0.0	0.5	V	1-4
J44	SCEA 3	76					
J45	SCEA 4	77					
J46	SCEA 5	78					
J47	SCEA 6	79					
J48	SCEA 7	80					
J49	SCEA 8	82					
J50	SCEA 9	83					
J51	SCEA 10	84					
J52	SCEA 11	85					
J53	SCEA 12	95					
J54	SCEA 13	96					
J55	SCEA 14	97					
J56	SCEA 15	98					
J57	MEM ADR REG	99					
J58	MEM DATA REG	100	1.6	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J58
Power supply current limit	3.27 A	NA	J1-J58
Forcing current, (Iol)	1.6 mA	± 160 uA	J1
	6.0 mA	± 600 uA	J2,J3
	1.6 mA	± 160 uA	J4-J27
	6.0 mA	± 600 uA	J28-J33
	1.6 mA	± 160 uA	J34
	6.0 mA	± 600 uA	J35
	1.6 mA	± 160 uA	J36,J37
	6.0 mA	± 600 uA	J38
	1.6 mA	± 160 uA	J39
	6.0 mA	± 600 uA	J40
	1.6 mA	± 160 uA	J41-J58
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	J1-J58
low, (VL)	0.0 V	± 100 mV	J1-J58
Output voltage measurement	NA	± 25 mV	J1-J58

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	PROM CLK	4	-18.0	-1.2	0.0	V	1-3
L2	SERIAL IN	8	-18.0	-1.5	0.0	V	1-3
L3	C4 CLK	34	-18.0	-1.2	0.0	V	1-3
L4	MA CLR	43					
L5	S/D CLK	52					
L6	LOAD SHIFT	53					
L7	MAINT CLK	54					
L8	MPC MODE	55	-18.0	-1.2	0.0	V	1-3
L9	COMM PNDNG	56	-18.0	-1.5	0.0	V	1-3
L10	SCE/DEST 15	57					
L11	SCE/DEST 14	58					
L12	SCE/DEST 13	59					
L13	SCE/DEST 12	61					
L14	SCE/DEST 11	62					
L15	SCE/DEST 10	63					
L16	SCE/DEST 9	64					
L17	SCE/DEST 8	65					
L18	MP TO MPC	66	-18.0	-1.5	0.0	V	1-3
L19	CL I	67	-18.0	-1.2	0.0	V	1-3
L20	CL II	68	-18.0	-1.2	0.0	V	1-3
L21	CL III	69	-18.0	-1.5	0.0	V	1-3

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TABLE XI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L22	CHAIN REQ	70	-18.0	-1.2	0.0	V	1-3
L23	OP STP/RN	71	-18.0	-1.5	0.0	V	1-3
L24	SCE/DEST 7	86					
L25	SCE/DEST 6	87					
L26	SCE/DEST 5	88					
L27	SCE/DEST 4	89					
L28	SCE/DEST 3	91					
L29	SCE/DEST 2	92					
L30	SCE/DEST 1	93					
L31	SCE/DEST 0	94	-18.0	-1.5	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L31
Power supply current limit	NA	NA	L1-L31
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L31

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	NA	NA	L1-L31
low, (VL)	NA	NA	L1-L31
Input voltage measurement	NA	± 60 mV	L1
	NA	± 75 mV	L2
	NA	± 60 mV	L3-L8
	NA	± 75 mV	L9-L18
	NA	± 60 mV	L19,L20
	NA	± 75 mV	L21
	NA	± 60 mV	L22
	NA	± 75 mV	L23-L31

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	4.7	8.8	uF	1-4
M2	PSFC	51	4.3	7.9	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 205 nF ± 180 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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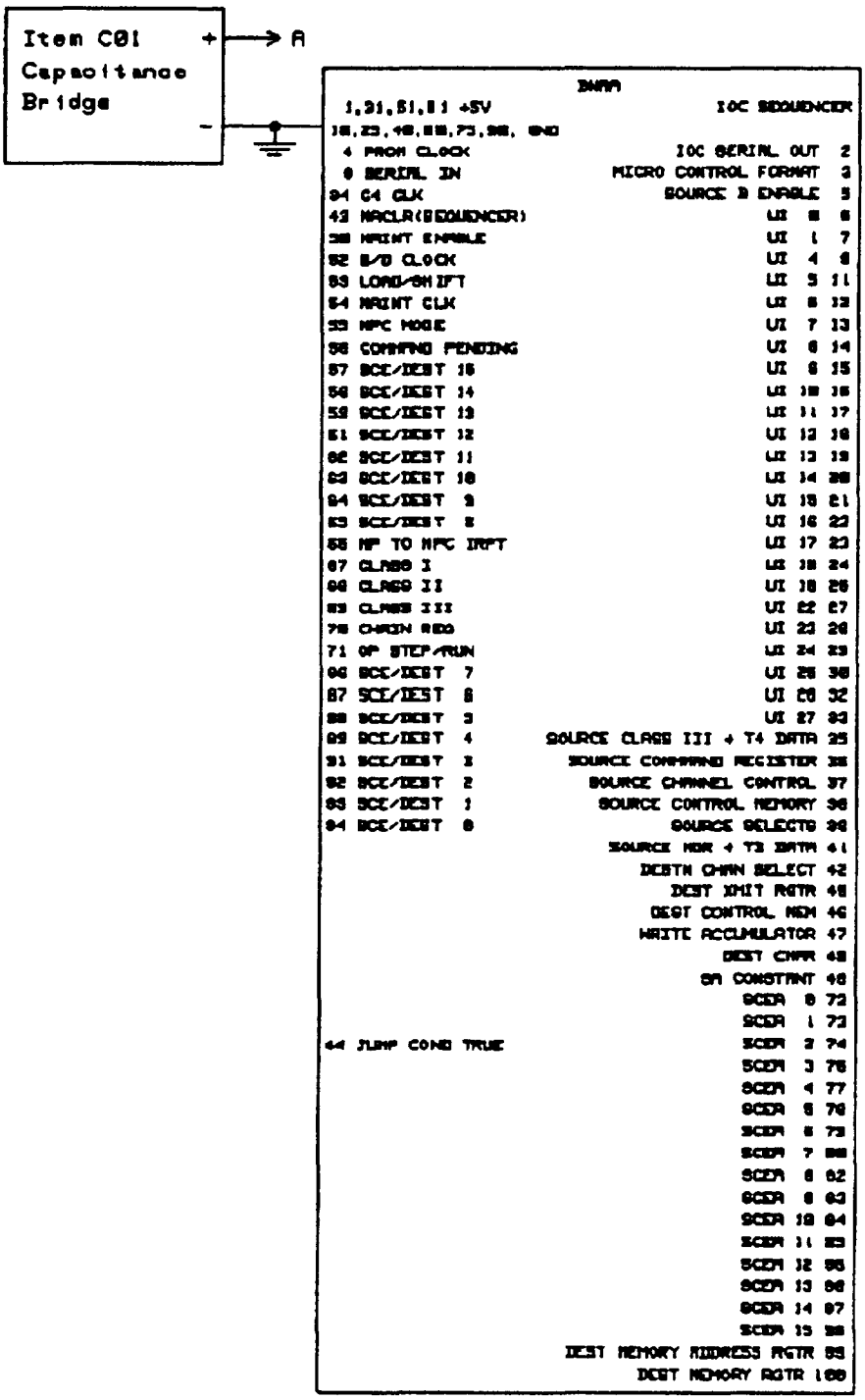


FIGURE 10. Power supply filter capacitance test circuit for table XII.

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TABLE XIII. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1	910	2180	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	3.27 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 63.5 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Functional test.

Test cond	Test description	Notes	Figs
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3	9,11
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3	9,11

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	3.27 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

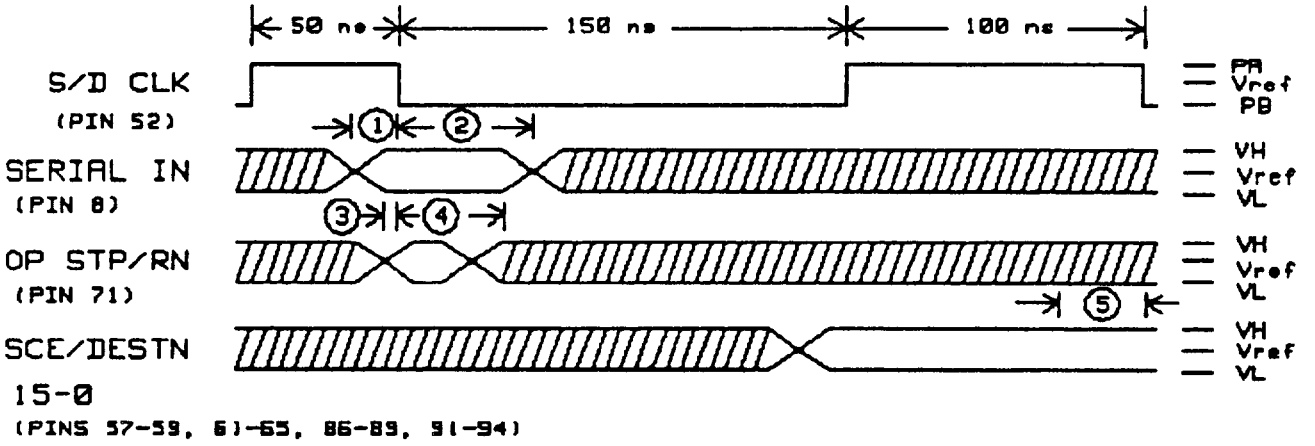
(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XIV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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NOTES:

1. (1) SERIAL IN to S/D CLOCK. Setup time: 14 nS minimum.

2. (2) S/D CLOCK to SERIAL IN. Hold time: 33 nS minimum.

3. (3) OP STP/RN to S/D CLK. Setup time: 4 nS minimum

4. (4) S/D CLK to OP STP/RN. Hold time: 33 nS minimum.

5. (5) SCE/DEST 15 - SCE/DEST 0 to S/D CLK. Setup time: 10 nS minimum.

FIGURE 10. Input timing parameters for table XIV.

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TABLE XV. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	3.27 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

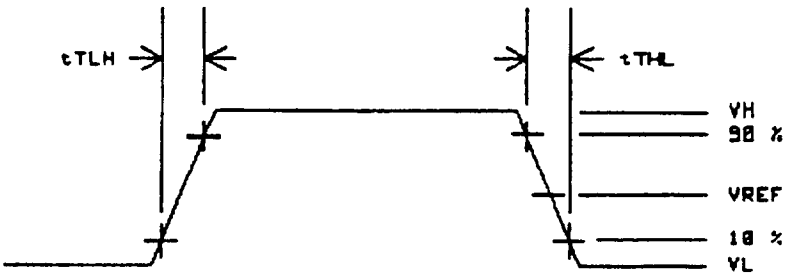
(b) The output compare levels shall be as follows:

High level: Vref ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ Vref

(c) The dynamic parameters are found in table XVI.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XV.

6. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	MAINT CLK to IOC SER OUT	-	-	-	137	137	137	nS	12
1	tpd	MAINT CLK to MIC CNT FMT	-	-	-	137	137	137	nS	12
1	tpd	MAINT CLK to DEST CHN SEL	-	-	-	137	137	137	nS	12
1	tpd	MAINT CLK to JMP CON TRU	-	-	-	137	137	137	nS	12
1	tpd	MAINT CLK to WRITE ACC	-	-	-	137	137	137	nS	12
2	tpd	S/D CLK to UI 0	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 1	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 4	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 5	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 6	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 7	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 8	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 9	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 10	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 11	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 12	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 13	-	-	-	82	82	82	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	S/D CLK to UI 14	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 15	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 16	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 17	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 18	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 19	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 22	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 23	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 24	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 25	-	-	-	82	82	82	nS	12
2	tpd	S/D CLK to UI 26	-	-	-	82	82	82	nS	12
2	tpd -	S/D CLK to UI 27	-	-	-	82	82	82	nS	12
3	tpd	C4 CLK to SC B ENBL	-	-	-	18	18	18	nS	12
4	tpd	MAINT CLK to C III + T4	-	-	-	80	80	80	nS	12
4	tpd	MAINT CLK to SRC COMM REG	-	-	-	80	80	80	nS	12
4	tpd	MAINT CLK to SRC CHAN CTL	-	-	-	80	80	80	nS	12
4	tpd	MAINT CLK to SRC CTL MEM	-	-	-	80	80	80	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	S/D CLK to DEST XMT REG	-	-	-	123	123	123	nS	12
5	tpd	S/D CLK to DEST CTL MEM	-	-	-	123	123	123	nS	12
5	tpd	S/D CLK to DEST CMAR	-	-	-	123	123	123	nS	12
5	tpd	S/D CLK to SA CNST	-	-	-	123	123	123	nS	12
5	tpd	S/D CLK to MEM ADR REG	-	-	-	123	123	123	nS	12
5	tpd	S/D CLK to MEM DATA REG	-	-	-	123	123	123	nS	12
6	tpd	S/D CLK to SCEA 0	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 1	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 2	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 3	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 4	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 5	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 6	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 7	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 8	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 9	-	-	-	81	81	81	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	S/D CLK to SCEA 10	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 11	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 12	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 13	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 14	-	-	-	81	81	81	nS	12
6	tpd	S/D CLK to SCEA 15	-	-	-	81	81	81	nS	12
7	tsu	SERIAL IN to S/D CLK	14	14	14	-	-	-	nS	12
8	th	S/D CLK to SERIAL IN	33	33	33	-	-	-	nS	12
9	tsu	OP STP/RN to S/D CLK	4	4	4	-	-	-	nS	12
10	th	S/D CLK to OP STP/RN	33	33	33	-	-	-	nS	12
11	tsu	SCE DEST 0 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 1 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 2 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 3 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 4 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 5 to S/D CLK	10	10	10	-	-	-	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tsu	SCE DEST 6 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 7 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 8 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 9 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 10 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 11 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 12 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 13 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 14 to S/D CLK	10	10	10	-	-	-	nS	12
11	tsu	SCE DEST 15 to S/D CLK	10	10	10	-	-	-	nS	12

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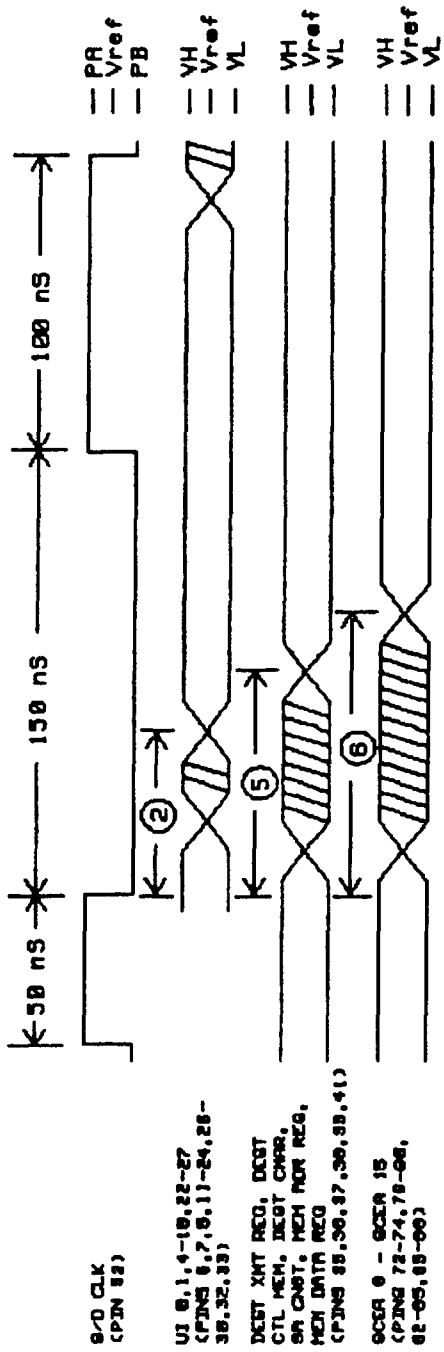


FIGURE 12a. Dynamic waveforms for table XVI.

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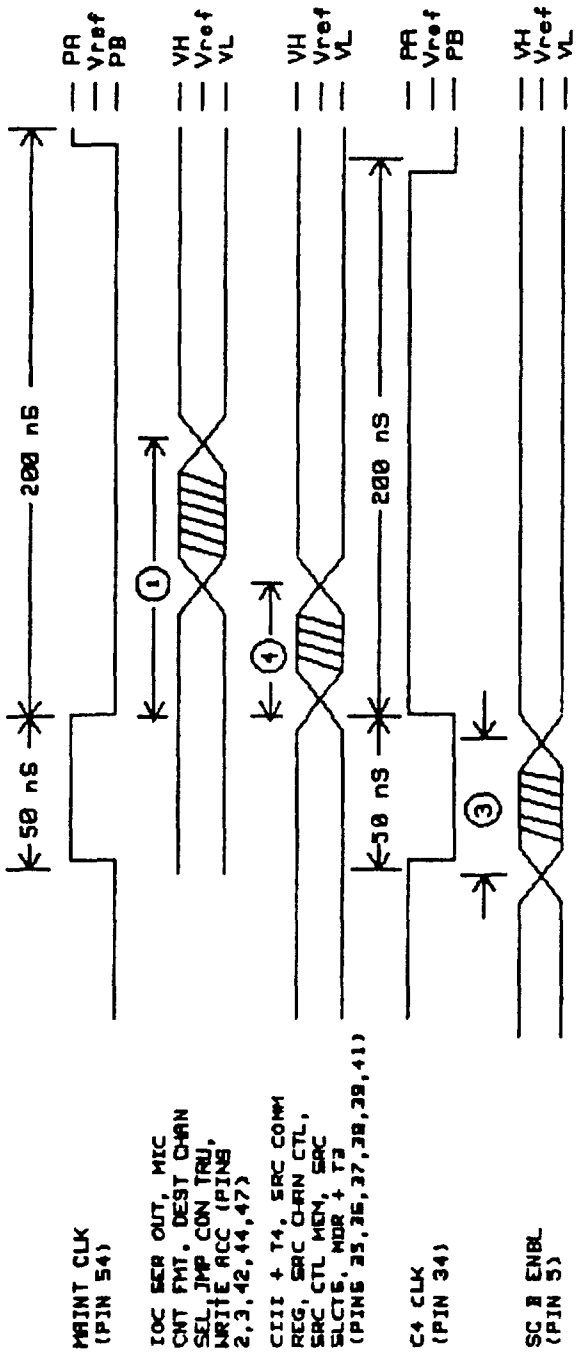


FIGURE 12b. Dynamic waveforms for table XVI.

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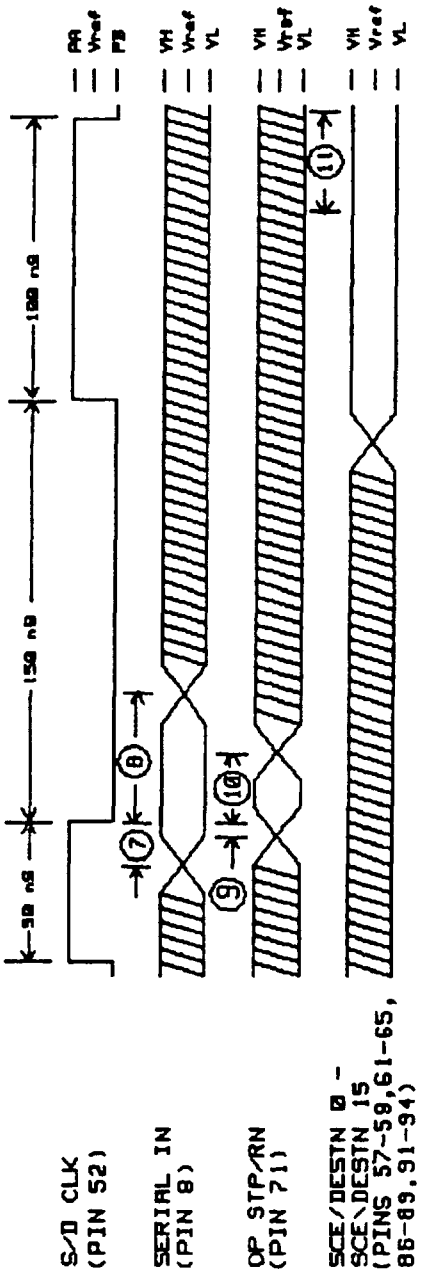


FIGURE 12c. Dynamic waveforms for table XVI.

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TABLE XVII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	PROM CLK	4	2	50	pF	1-4
R2	SERIAL IN	8	2	75	pF	1-4
R3	C4 CLK	34	2	100	pF	1-4
R4	MA CLR	43	24	321	pF	1-4
R5	MAINT ENBL	50	2	175	pF	1-4
R6	S/D CLK	52	4	100	pF	1-4
R7	LOAD SHIFT	53	4	75	pF	1-4
R8	MAINT CLK	54	4	75	pF	1-4
R9	MPC MODE	55	2	75	pF	1-4
R10	COMM PNDNG	56	4	75	pF	1-4
R11	SCE/DEST 15	57	2	50	pF	1-4
R12	SCE/DEST 14	58				
R13	SCE/DEST 13	59				
R14	SCE/DEST 12	61				
R15	SCE/DEST 11	62				
R16	SCE/DEST 10	63				
R17	SCE/DEST 9	64				
R18	SCE/DEST 8	65	2	50	pF	1-4
R19	MP TO MPC	66	2	75	pF	1-4
R20	CL I	67	4	75	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	CL II	68	4	75	pF	1-4
R22	CL III	69	2	75	pF	1-4
R23	CHAIN REQ	70	8	100	pF	1-4
R24	OP STP/RN	71	2	50	pF	1-4
R25	SCE/DEST 7	86				
R26	SCE/DEST 6	87				
R27	SCE/DEST 5	88				
R28	SCE/DEST 4	89				
R29	SCE/DEST 3	91				
R30	SCE/DEST 2	92				
R31	SCE/DEST 1	93				
R32	SCE/DEST 0	94	2	50	pF	1-4
Outputs						
R33	UI 0	6	6	200	pF	1-4
R34	UI 1	7	6	200	pF	1-4
R35	UI 4	9	6	150	pF	1-4
R36	UI 5	11				
R37	UI 6	12	6	150	pF	1-4
R38	UI 7	13	8	175	pF	1-4
R39	UI 8	14	4	75	pF	1-4
R40	UI 9	15	6	100	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	UI 10	16	6	100	pF	1-4
R42	UI 11	17	4	100	pF	1-4
R43	UI 12	18	4	75	pF	1-4
R44	UI 13	19	4	100	pF	1-4
R45	UI 14	20				
R46	UI 15	21	4	100	pF	1-4
R47	UI 16	22	4	75	pF	1-4
R48	UI 17	23				
R49	UI 18	24	4	75	pF	1-4
R50	UI 19	26	4	100	pF	1-4
R51	UI 22	27	2	50	pF	1-4
R52	UI 23	28				
R53	UI 24	29				
R54	UI 25	30				
R55	UI 26	32				
R56	UI 27	33	2	50	pF	1-4
R57	SCEA 0	72	4	75	pF	1-4
R58	SCEA 1	73				
R59	SCEA 2	74				
R60	SCEA 3	76				
R61	SCEA 4	77	4	75	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R62	SCEA 5	78	4	75	pF	1-4
R63	SCEA 6	79				
R64	SCEA 7	80				
R65	SCEA 8	82				
R66	SCEA 9	83				
R67	SCEA 10	84				
R68	SCEA 11	85	4	75	pF	1-4
R69	SCEA 12	95	2	50	pF	1-4
R70	SCEA 13	96	4	75	pF	1-4
R71	SCEA 14	97	2	50	pF	1-4
R72	SCEA 15	98	2	50	pF	1-4
R73	JMP CON TRU	44	6	200	pF	1-4

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R73
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R73

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 2.4 pF	R1
	NA	± 3.65 pF	R2
	NA	± 4.9 pF	R3
	NA	± 14.8 pF	R4
	NA	± 8.65 pF	R5
	NA	± 4.8 pF	R6
	NA	± 3.55 pF	R7,R8
	NA	± 3.65 pF	R9
	NA	± 3.55 pF	R10
	NA	± 2.4 pF	R11-R18
	NA	± 3.65 pF	R19
	NA	± 3.55 pF	R20,R21
	NA	± 3.65 pF	R22
	NA	± 4.6 pF	R23
	NA	± 2.4 pF	R24-R32
	NA	± 9.7 pF	R33,R34
	NA	± 7.2 pF	R35-R37
	NA	± 8.35 pF	R38
	NA	± 3.55 pF	R39
	NA	± 4.7 pF	R40,R41
	NA	± 4.8 pF	R42
	NA	± 3.55 pF	R43
	NA	± 4.8 pF	R44-R46
	NA	± 3.55 pF	R47-R49
	NA	± 4.8 pF	R50
	- NA	± 2.4 pF	R51-R56
	NA	± 3.55 pF	R57-R68
	NA	± 2.4 pF	R69
	NA	± 3.55 pF	R70
	NA	± 2.4 pF	R71,R72
	NA	± 9.7 pF	R73

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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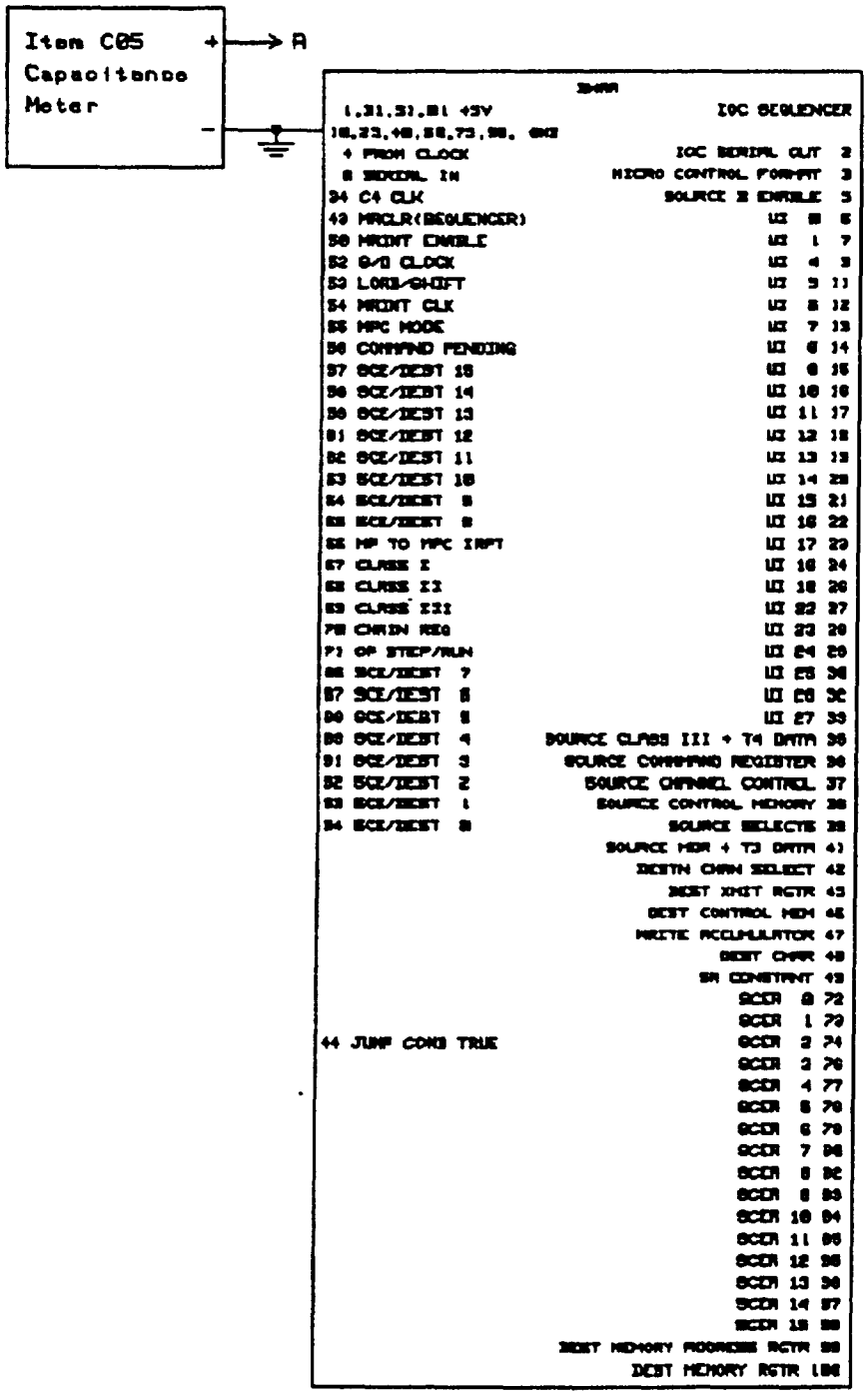


FIGURE 13. Input and output capacitance test circuit for table XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	PSTA	1	0	400	mV	1-4	14,15
T2	PSTA	51	0	400	mV	1-4	14,15

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	3.27 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XVIII.

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NOTES: - Continued.

(c) Vcc pins 1 and 31 shall be shorted together, Vcc pins 51 and 81 shall be shorted together, and the scope point, probe 'A', shall be connected to the pin indicated in table XVIII.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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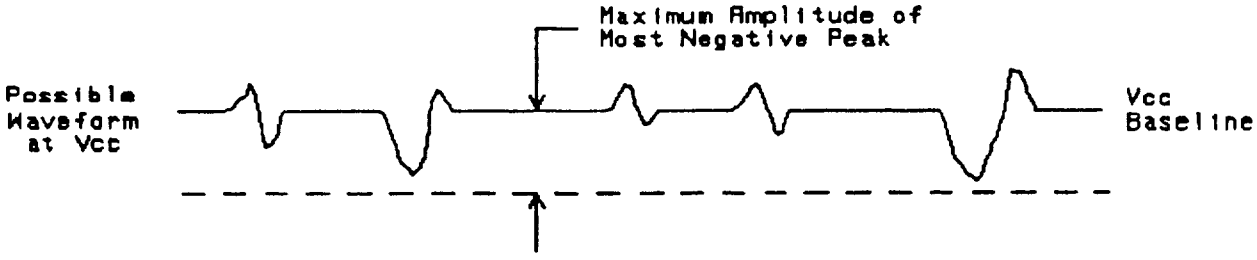
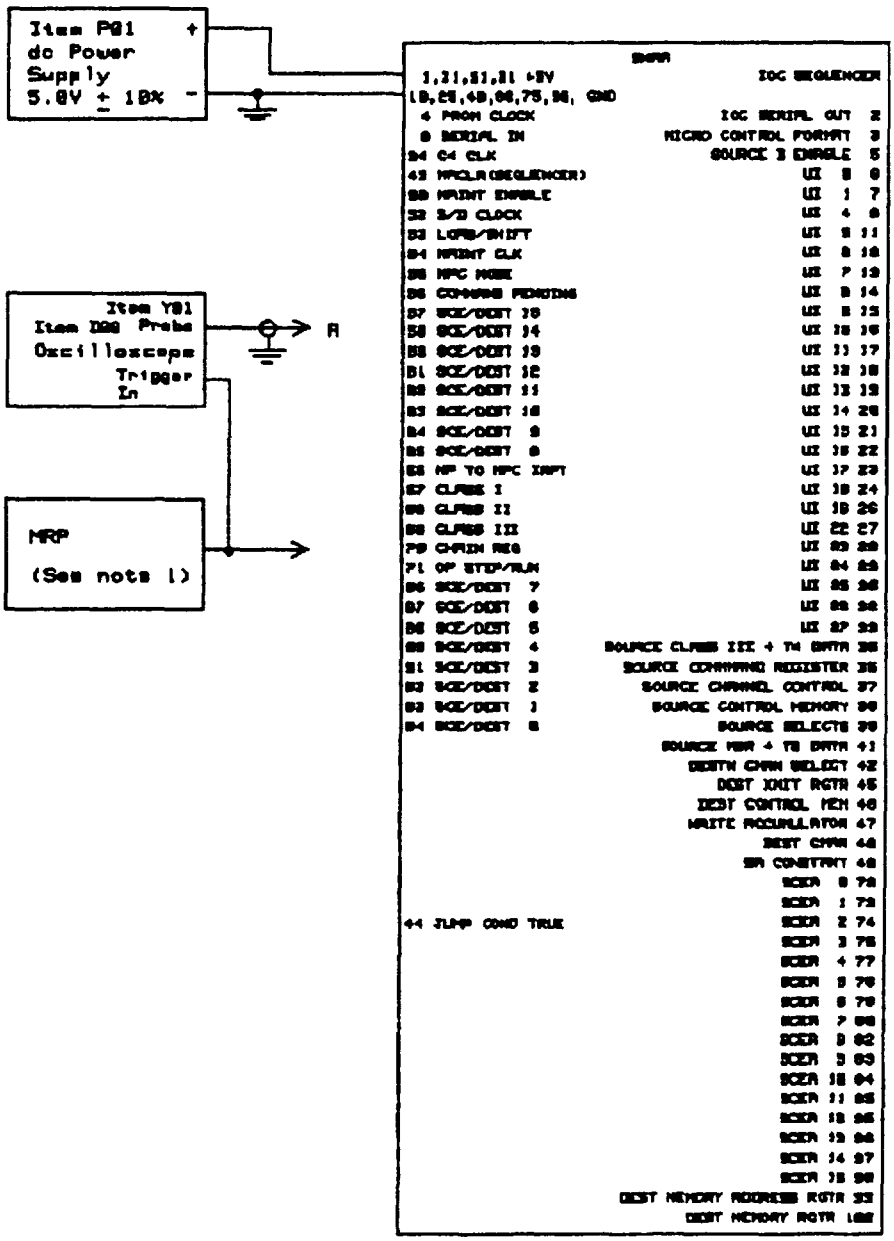


FIGURE 14. Power supply transient amplitude waveform for table XVIII.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XVIII.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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MIL-M-28787/507	Modules, Standard Electronic, IOC Sequencer Module, Key Code BWAA

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5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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